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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Rotational Motion

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	189 questions
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Answer key	at the end

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MCQ (Single Correct Answer) 189 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $t = 0$ the cylinder has a translational velocity $v_o = 49\ \mathrm{m/s}$, perpendicular to its axis and a rotational velocity $v_o/4R$ about the centre. The time taken by the cylinder to start rolling is ____ seconds. (coefficient of kinetic friction $\mu_K = 0.25$ and $g = 9.8\ \mathrm{m/s^2}$)

Open solution

(A) 15 (B) 5 (C) 10 (D) 7.5

Q2. JEE Main 2026 (Online) 6th April Evening Shift

A solid sphere (A) of mass $5m$ and a spherical shell (B) of mass m, both having same radius, are placed on a rough surface. When a force of same magnitude is applied tangentially at the highest points of A and B, they start rolling without slipping with an acceleration of a_A and a_B, respectively. The ratio of a_A and a_B is ____ .

Open solution

(A) $5 : 21$ (B) $6 : 10$ (C) $21 : 25$ (D) $1 : 5$

Q3. JEE Main 2026 (Online) 6th April Morning Shift

A solid sphere of radius 4 cm and mass 5 kg is rotating (rotation axis is passing through the centre of the sphere) with an angular velocity of 1200 rpm . It is brought to rest in 10 s by applying a constant torque. The torque applied and the number of rotations it made before it comes to rest are ____ and ____ respectively.

Open solution

(A) $0.128\pi\mathrm{Nm}, 100$ (B) $0.0128\pi\mathrm{Nm}, 50$ (C) $0.128\pi\mathrm{Nm}, 50$ (D) $0.0128\pi\mathrm{Nm}, 100$

Q4. JEE Main 2026 (Online) 5th April Evening Shift

A wheel initially at rest is subjected to a uniform angular acceleration about its axis. In the first 2 s it rotates through an angle θ_1 and in the next 2 s it rotates through an angle θ_2. The ratio $\frac{\theta_2}{\theta_1}$ is ____ .

Open solution

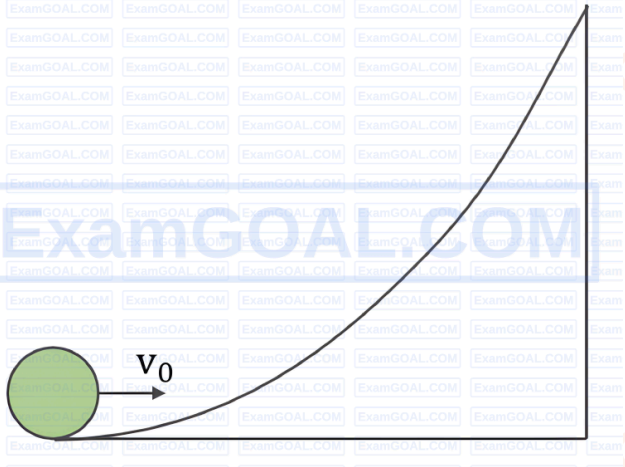
(A) 6 (B) 3 (C) 4 (D) $\frac{1}{3}$

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Q5. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

An object of uniform density rolls up the curved path with the initial velocity v_o as shown in the figure. If the maximum height attained by an object is $\frac{7v_0^2}{10\,g}$ (g = acceleration due to gravity), the object is a ____ .



(A) solid cylinder (B) ring (C) disc (D) solid sphere

Q6. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

A solid sphere of mass M and radius R is divided into two unequal parts. The smaller part having mass $M/8$ is converted into a sphere of radius r and the larger part is converted into a circular disc of thickness t and radius $2R$. If I_1 is moment of inertia of a sphere having radius r about an axis through its centre and I_2 is the moment of inertia of a disc about its diameter, the ratio of their moment of inertia $I_2/I_1 =$ ____

(A) 35 (B) 70 (C) 140 (D) 210

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = \left(10t^2\hat{i} + 5t^3\hat{j}\right)$ m. At $t = 1$ s, which of the following statements are **correct**?

A. The linear momentum $\vec{p} = \left(2\hat{i} + 1.5\hat{j}\right)$ kg·m/s.

B. The force acting on the object $\vec{F} = \left(2\hat{i} + 3\hat{j}\right)$ N.

C. The angular momentum of the object about its origin $\vec{L} = 15\hat{k}$ J·s.

D. The torque acting on the object about its origin $\vec{\tau} = 20\hat{k}$ N·m.

Choose the **correct** answer from the options given below:

(A) A, B and C only (B) B, C and D only (C) A, C and D only (D) A, B and D only

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Q8. JEE Main 2026 (Online) 28th January Evening Shift

When the position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ changes sign as $-\vec{r}$, which one of the following vector will not flip under sign change?

Open solution

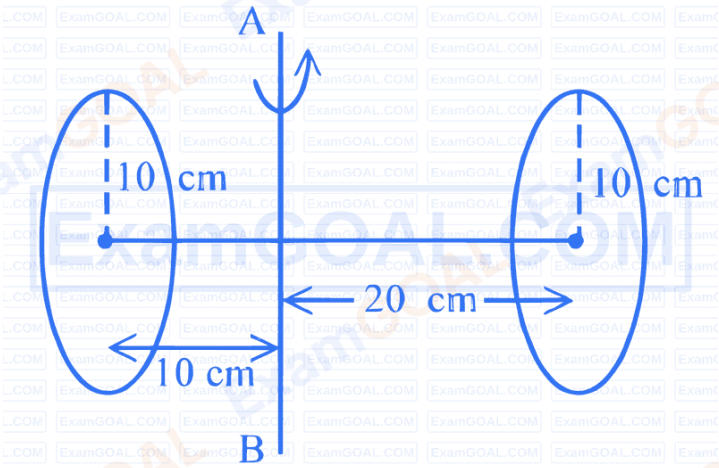
(A) Velocity
(B) Linear momentum
(C) Acceleration
(D) Angular momentum

Q9. JEE Main 2026 (Online) 28th January Morning Shift

Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure.

If the mass of each disc is 600 gm and applied torque between two discs is 43×10^5 dyne.cm, the angular acceleration of the discs about the given axis AB is ____ rad/s^2.

Open solution



(A) 100
(B) 22
(C) 27
(D) 11

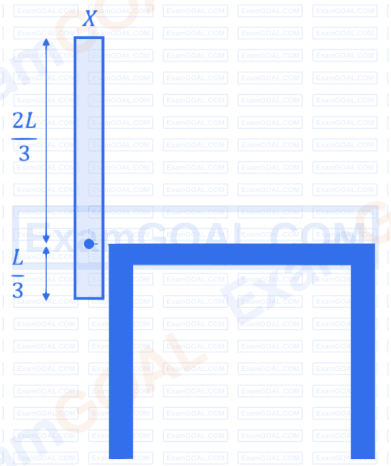
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Q10. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is ____ .

($g =$ gravitational acceleration)



(A) $\sqrt{\frac{3}{2}\frac{g}{L}}$

(B) $\sqrt{\frac{3g}{L}}$

(C) $\frac{3}{\sqrt{2}}\sqrt{\frac{g}{L}}$

(D) $\frac{1}{\sqrt{2}}\sqrt{\frac{g}{L}}$

Q11. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Two masses 400 g and 350 g are suspended from the ends of a light string passing over a heavy pulley of radius 2 cm . When released from rest the heavier mass is observed to fall 81 cm in 9 s . The rotational inertia of the pulley is ____ $\mathrm{kg \cdot m^2}$. $\left(\mathrm{g} = 9.8\ \mathrm{m/s^2}\right)$

(A) 8.3×10^{-3}

(B) 4.75×10^{-3}

(C) 1.86×10^{-2}

(D) 9.5×10^{-3}

Q12. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Two small balls with masses m and 2 m are attached to both ends of a rigid rod of length d and negligible mass. If angular momentum of this system is L about an axis (A) passing through its centre of mass and perpendicular to the rod then angular velocity of the system about A is :

(A) $\frac{4}{3}\frac{L}{md^2}$

(B) $\frac{3}{2}\frac{L}{md^2}$

(C) $\frac{2L}{5md^2}$

(D) $\frac{2L}{md^2}$

Q13. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The moment of inertia of a square loop made of four uniform solid cylinders, each having radius R and length $L(\mathrm{R} < \mathrm{L})$ about an axis passing through the mid points of opposite sides, is (Take the mass of the entire loop as M) :

(A) $\frac{3}{4}MR^2 + \frac{1}{6}ML^2$

(B) $\frac{3}{8}MR^2 + \frac{7}{12}ML^2$

(C) $\frac{3}{4}MR^2 + \frac{7}{12}ML^2$

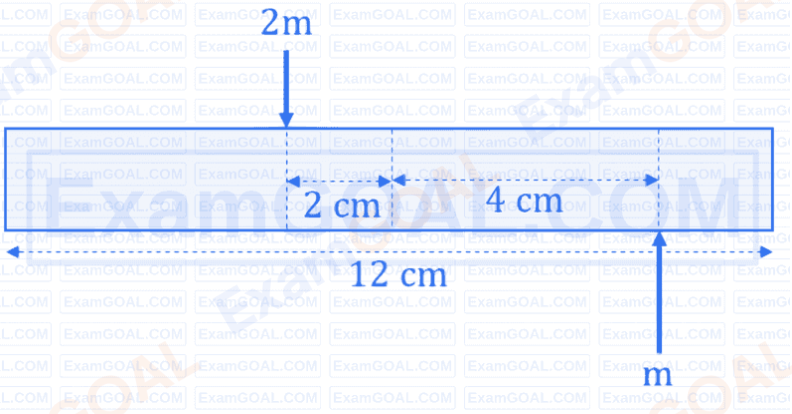
(D) $\frac{3}{8}MR^2 + \frac{1}{6}ML^2$

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Q14. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

A uniform bar of length 12 cm and mass 20 m lies on a smooth horizontal table. Two point masses m and $2m$ are moving in opposite directions with same speed of v and in the same plane as the bar, as shown in figure. These masses strike the bar simultaneously and get stuck to it. After collision the entire system is rotating with angular frequency ω. The ratio of v and ω is :

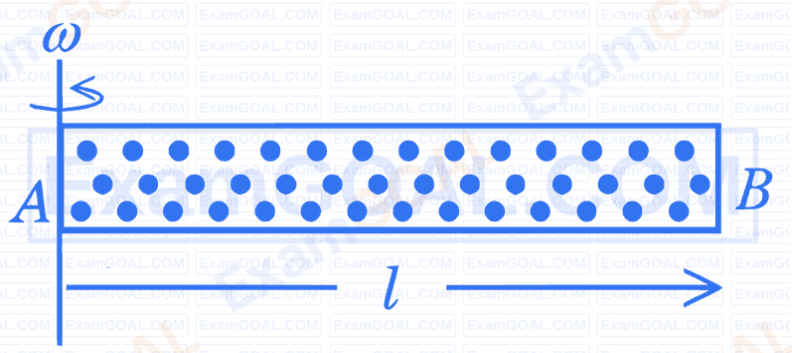


(A) 33 (B) $2\sqrt{88}$ (C) 32 (D) 66

Q15. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

A cylindrical tube AB of length l, closed at both ends contains an ideal gas of 1 mol having molecular weight M. The tube is rotated in a horizontal plane with constant angular velocity ω about an axis perpendicular to AB and passing through the edge at end A, as shown in the figure. If P_A and P_B are the pressures at A and B respectively, then (Consider the temperature is same at all points in the tube)



(A) $P_B = P_A e^{\left(\frac{M\omega^2 l^2}{3RT}\right)}$ (B) $P_B = P_A$

(C) $P_B = P_A e^{\left(\frac{M\omega^2 l^2}{2RT}\right)}$ (D) $P_B = P_A e^{\left(\frac{M\omega^2 l^2}{RT}\right)}$

Q16. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

A solid sphere of mass 5 kg and radius 10 cm is kept in contact with another solid sphere of mass 10 kg and radius 20 cm . The moment of inertia of this pair of spheres about the tangent passing through the point of contact is ____ $\mathrm{kg \cdot m^2}$.

(A) 0.18 (B) 0.72 (C) 0.36 (D) 0.63

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Q17. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

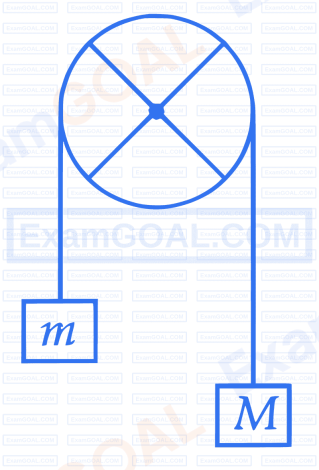
Two cars **A** and **B** each of mass 10^3 kg are moving on parallel tracks separated by a distance of 10 m, in same direction with speeds 72 km/h and 36 km/h. The magnitude of angular momentum of car **A** with respect to car **B** is _____ J·s.

(A) 3.6×10^5 (B) 10^5 (C) 3×10^5 (D) 2×10^5

Q18. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

The pulley shown in figure is made using a thin rim and two rods of length equal to diameter of the rim. The rim and each rod have a mass of **M**. Two blocks of mass of **M** and **m** are attached to two ends of a light string passing over the pulley, which is hinged to rotate freely in vertical plane about its center. The magnitudes of the acceleration experienced by the blocks is _____ (assume no slipping of string on pulley).



(A) $\frac{(M-m)g}{\left[\left(\frac{8}{3}\right)M+m\right]}$ (B) $\frac{(M-m)g}{2M+m}$

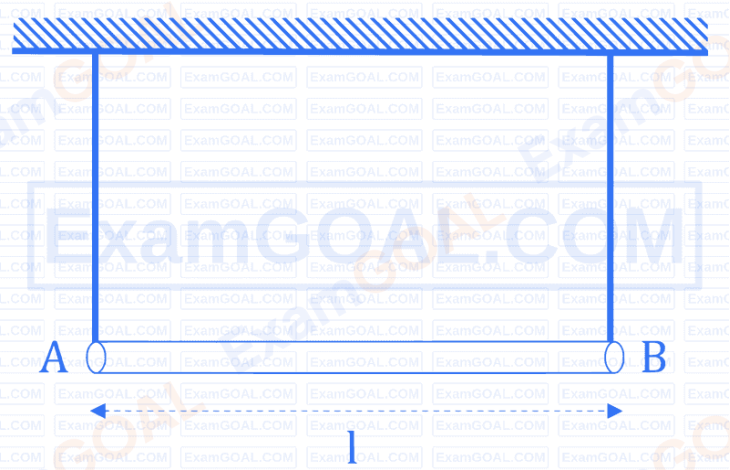
(C) $\frac{(M-m)g}{M+m}$ (D) $\frac{(M-m)g}{\left[\left(\frac{13}{6}\right)M+m\right]}$

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Q19. JEE Main 2026 (Online) 21st January Morning Shift

A uniform rod of mass m and length l is suspended by means of two identical inextensible light strings as shown in figure. Tension in one string immediately after the other string is cut, is ____ . (g acceleration due to gravity)

Open solution



(A) $\mathrm{mg/s}$ (B) $mg/4$ (C) mg (D) $mg/2$

Q20. JEE Main 2025 (Online) 8th April Evening Shift

A rod of linear mass density 'λ' and length 'L' is bent to form a ring of radius 'R'. Moment of inertia of ring about any of its diameter is.

Open solution

(A) $\frac{\lambda L^3}{8\pi^2}$ (B) $\frac{\lambda L^3}{16\pi^2}$ (C) $\frac{\lambda L^3}{4\pi^2}$ (D) $\frac{\lambda L^3}{12}$

Q21. JEE Main 2025 (Online) 4th April Evening Shift

A wheel is rolling on a plane surface. The speed of a particle on the highest point of the rim is $8\ \mathrm{m/s}$. The speed of the particle on the rim of the wheel at the same level as the centre of wheel, will be :

Open solution

(A) $4\sqrt{2}\ \mathrm{m/s}$ (B) $8\ \mathrm{m/s}$
(C) $4\ \mathrm{m/s}$ (D) $8\sqrt{2}\ \mathrm{m/s}$

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Q22. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Which of the following are correct expression for torque acting on a body?

A. $\vec{\tau} = \vec{r} \times \vec{L}$

B. $\vec{\tau} = \frac{d}{dt}(\vec{r} \times \vec{p})$

C. $\vec{\tau} = \vec{r} \times \frac{d\vec{p}}{dt}$

D. $\vec{\tau} = I\vec{\alpha}$

E. $\vec{\tau} = \vec{r} \times \vec{F}$

($\vec{r}$ = position vector; $\vec{p}$ = linear momentum; $\vec{L}$ = angular momentum; $\vec{\alpha}$ = angular acceleration; I = moment of inertia; $\vec{F}$ = force; t = time)

Choose the correct answer from the options given below:

(A) A, B, D and E Only

(B) C and D Only

(C) B, C, D and E Only

(D) B, D and E Only

Q23. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

If $\vec{L}$ and $\vec{P}$ represent the angular momentum and linear momentum respectively of a particle of mass ' m ' having position vector as $\vec{r} = a(\hat{i}\cos\omega t + \hat{j}\sin\omega t)$. The direction of force is

(A) Opposite to the direction of $\vec{L}$

(B) Opposite to the direction of $\vec{L} \times \vec{P}$

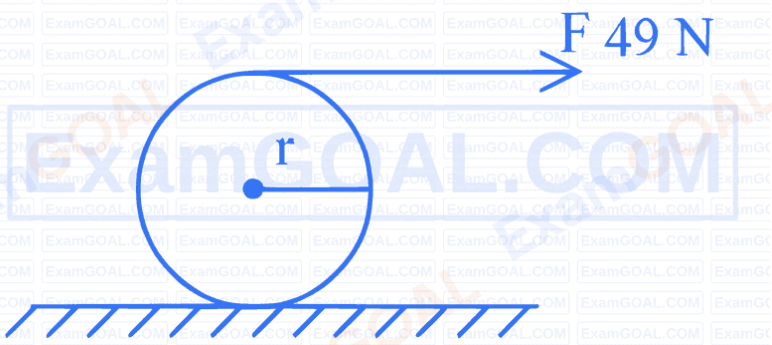
(C) Opposite to the direction of $\vec{r}$

(D) Opposite to the direction of $\vec{P}$

Q24. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

A force of 49 N acts tangentially at the highest point of a sphere (solid) of mass 20 kg , kept on a rough horizontal plane. If the sphere rolls without slipping, then the acceleration of the center of the sphere is



(A) $2.5\ \mathrm{m/s^2}$

(B) $3.5\ \mathrm{m/s^2}$

(C) $0.25\ \mathrm{m/s^2}$

(D) $0.35\ \mathrm{m/s^2}$

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Q25. JEE Main 2025 (Online) 2nd April Evening Shift

The moment of inertia of a circular ring of mass M and diameter r about a tangential axis lying in the plane of the ring is :

Open solution

(A) $\frac{3}{8}Mr^2$ (B) $2Mr^2$ (C) $\frac{1}{2}Mr^2$ (D) $\frac{3}{2}Mr^2$

Q26. JEE Main 2025 (Online) 2nd April Morning Shift

Moment of inertia of a rod of mass ' M ' and length ' L ' about an axis passing through its center and normal to its length is ' α '. Now the rod is cut into two equal parts and these parts are joined symmetrically to form a cross shape. Moment of inertia of cross about an axis passing through its center and normal to plane containing cross is :

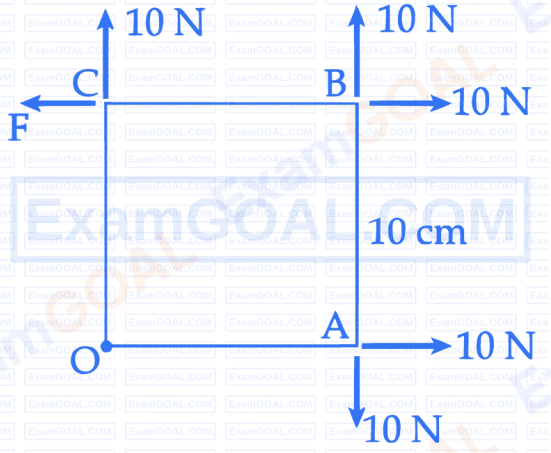
Open solution

(A) $\alpha/4$ (B) $\alpha/8$ (C) α (D) $\alpha/2$

Q27. JEE Main 2025 (Online) 2nd April Morning Shift

A square Lamina OABC of length 10 cm is pivoted at ' O '. Forces act at Lamina as shown in figure. If Lamina remains stationary, then the magnitude of F is :

Open solution



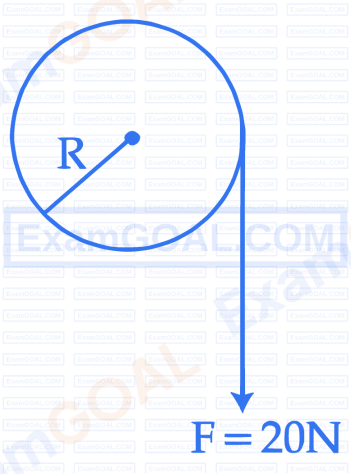
(A) 10 N (B) 0 (zero) (C) $10\sqrt{2}$ N (D) 20 N

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Q28. JEE Main 2025 (Online) 2nd April Morning Shift

A cord of negligible mass is wound around the rim of a wheel supported by spokes with negligible mass. The mass of wheel is 10 kg and radius is 10 cm and it can freely rotate without any friction. Initially the wheel is at rest. If a steady pull of 20 N is applied on the cord, the angular velocity of the wheel, after the cord is unwound by 1 m , would be:

Open solution



(A) 20 rad/s (B) 30 rad/s (C) 10 rad/s (D) 0 rad/s

Q29. JEE Main 2025 (Online) 28th January Evening Shift

A uniform rod of mass 250 g having length 100 cm is balanced on a sharp edge at 40 cm mark. A mass of 400 g is suspended at 10 cm mark. To maintain the balance of the rod, the mass to be suspended at 90 cm mark, is

Open solution

(A) 290 g (B) 200 g (C) 190 g (D) 300 g

Q30. JEE Main 2025 (Online) 24th January Evening Shift

A solid sphere and a hollow sphere of the same mass and of same radius are rolled on an inclined plane. Let the time taken to reach the bottom by the solid sphere and the hollow sphere be t_1 and t_2, respectively, then

Open solution

(A) $t_1 < t_2$ (B) $t_1 = 2t_2$ (C) $t_1 > t_2$ (D) $t_1 = t_2$

Q31. JEE Main 2025 (Online) 24th January Evening Shift

A solid sphere is rolling without slipping on a horizontal plane. The ratio of the linear kinetic energy of the centre of mass of the sphere and rotational kinetic energy is :

Open solution

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{5}{2}$ (D) $\frac{2}{5}$

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Q32. JEE Main 2025 (Online) 24th January Morning Shift

A uniform solid cylinder of mass ' m ' and radius ' r ' rolls along an inclined rough plane of inclination 45°. If it starts to roll from rest from the top of the plane then the linear acceleration of the cylinder's axis will be

Open solution

(A) $\sqrt{2}\,g$ (B) $\frac{1}{\sqrt{2}}\,g$ (C) $\frac{1}{3\sqrt{2}}\,g$ (D) $\frac{\sqrt{2}g}{3}$

Q33. JEE Main 2025 (Online) 23rd January Evening Shift

A circular disk of radius R meter and mass M kg is rotating around the axis perpendicular to the disk. An external torque is applied to the disk such that $\theta(t) = 5t^2 - 8t$, where $\theta(t)$ is the angular position of the rotating disc as a function of time t. How much power is delivered by the applied torque, when $t = 2\,s$?

Open solution

(A) $60MR^2$ (B) $72MR^2$ (C) $8MR^2$ (D) $108MR^2$

Q34. JEE Main 2025 (Online) 23rd January Morning Shift

A solid sphere of mass ' m ' and radius ' r ' is allowed to roll without slipping from the highest point of an inclined plane of length ' L ' and makes an angle 30° with the horizontal. The speed of the particle at the bottom of the plane is v_1. If the angle of inclination is increased to 45° while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2. The ratio $v_1^2 : v_2^2$ is

Open solution

(A) $1:3$ (B) $1:2$ (C) $1:\sqrt{2}$ (D) $1:\sqrt{3}$

Q35. JEE Main 2025 (Online) 22nd January Evening Shift

The torque due to the force $(2\hat{i} + \hat{j} + 2\hat{k})$ about the origin, acting on a particle whose position vector is $(\hat{i} + \hat{j} + \hat{k})$, would be

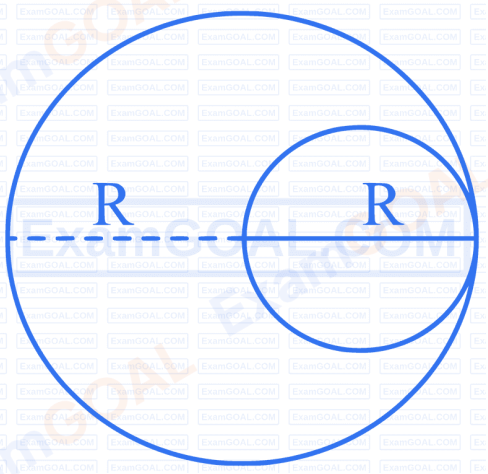
Open solution

(A) $\hat{j} + \hat{k}$ (B) $\hat{i} - \hat{k}$ (C) $\hat{i} - \hat{j} + \hat{k}$ (D) $\hat{i} + \hat{k}$

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Q36. JEE Main 2025 (Online) 22nd January Morning Shift

A uniform circular disc of radius ' R ' and mass ' M ' is rotating about an axis perpendicular to its plane and passing through its centre. A small circular part of radius $R/2$ is removed from the original disc as shown in the figure. Find the moment of inertia of the remaining part of the original disc about the axis as given above.



Open solution

(A) $\frac{17}{32}\mathrm{MR}^2$ (B) $\frac{13}{32}\mathrm{MR}^2$ (C) $\frac{9}{32}\mathrm{MR}^2$ (D) $\frac{7}{32}\mathrm{MR}^2$

Q37. JEE Main 2024 (Online) 9th April Morning Shift

A heavy iron bar, of weight W is having its one end on the ground and the other on the shoulder of a person. The bar makes an angle θ with the horizontal. The weight experienced by the person is :

Open solution

(A) $W\sin\theta$ (B) W (C) $\frac{W}{2}$ (D) $W\cos\theta$

Q38. JEE Main 2024 (Online) 8th April Evening Shift

A thin circular disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with angular velocity ω. If another disc of same dimensions but of mass $\mathrm{M}/2$ is placed gently on the first disc co-axially, then the new angular velocity of the system is :

Open solution

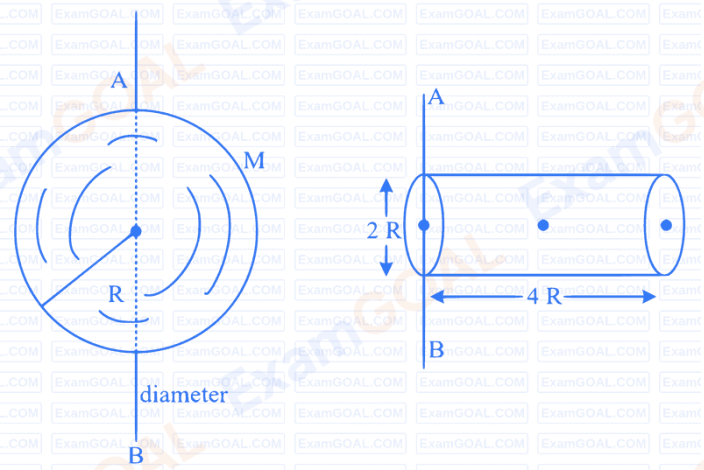
(A) $\frac{4}{5}\omega$ (B) $\frac{5}{4}\omega$ (C) $\frac{3}{2}\omega$ (D) $\frac{2}{3}\omega$

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Q39. JEE Main 2024 (Online) 5th April Morning Shift

Ratio of radius of gyration of a hollow sphere to that of a solid cylinder of equal mass, for moment of Inertia about their diameter axis AB as shown in figure is $\sqrt{8/x}$. The value of x is :

Open solution



(A) 34 (B) 51 (C) 67 (D) 17

Q40. JEE Main 2024 (Online) 1st February Evening Shift

A disc of radius R and mass M is rolling horizontally without slipping with speed v. It then moves up an inclined smooth surface as shown in figure. The maximum height that the disc can go up the incline is :

Open solution



(A) $\frac{3}{4}\frac{v^2}{g}$ (B) $\frac{v^2}{g}$

(C) $\frac{2}{3}\frac{v^2}{g}$ (D) $\frac{1}{2}\frac{v^2}{g}$

Q41. JEE Main 2024 (Online) 30th January Morning Shift

A particle of mass m is projected with a velocity 'u' making an angle of 30° with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height h is :

Open solution

(A) $\frac{mu^3}{\sqrt{2}g}$ (B) zero

(C) $\frac{\sqrt{3}}{2}\frac{mu^2}{g}$ (D) $\frac{\sqrt{3}}{16}\frac{mu^3}{g}$

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Q42. JEE Main 2024 (Online) 27th January Evening Shift

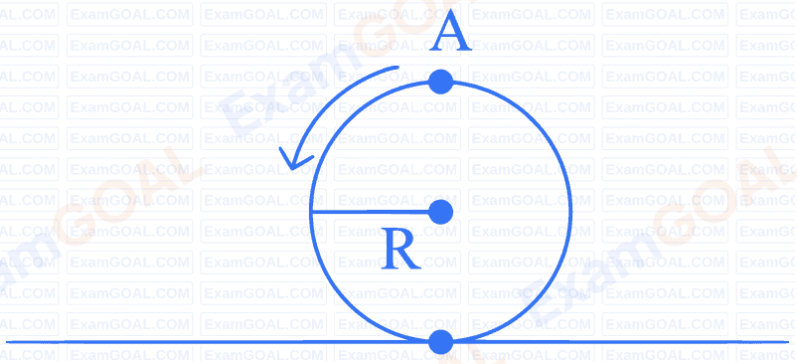
A heavy iron bar of weight $12\,\text{kg}$ is having its one end on the ground and the other on the shoulder of a man. The rod makes an angle 60° with the horizontal, the weight experienced by the man is :

(A) $3\,\text{kg}$ (B) $6\,\text{kg}$ (C) $6\sqrt{3}\,\text{kg}$ (D) $12\,\text{kg}$

Open solution

Q43. JEE Main 2023 (Online) 13th April Morning Shift

A disc is rolling without slipping on a surface. The radius of the disc is R. At $t = 0$, the top most point on the disc is A as shown in figure. When the disc completes half of its rotation, the displacement of point A from its initial position is



(A) $R\sqrt{(\pi^2+1)}$ (B) $2R$ (C) $R\sqrt{(\pi^2+4)}$ (D) $2R\sqrt{(1+4\pi^2)}$

Open solution

Q44. JEE Main 2023 (Online) 10th April Evening Shift

Given below are two statements: one is labelled as Assertion **A** and the other is labelled as Reason **R**

Assertion A : An electric fan continues to rotate for some time after the current is switched off.

Reason R : Fan continues to rotate due to inertia of motion.

In the light of above statements, choose the most appropriate answer from the options given below.

(A) A is not correct but R is correct

(B) A is correct but R is not correct

(C) Both A and R are correct and R is the correct explanation of A

(D) Both A and R are correct but R is NOT the correct explanation of A

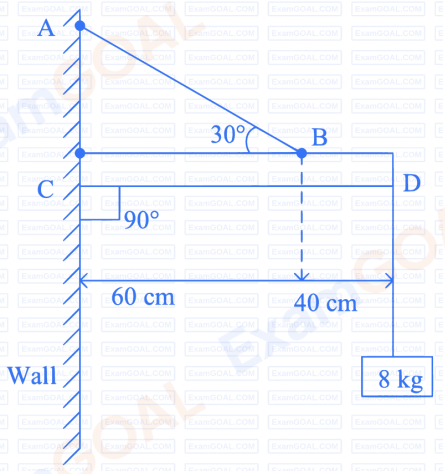
Open solution

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Q45. JEE Main 2023 (Online) 25th January Morning Shift

An object of mass 8 kg is hanging from one end of a uniform rod CD of mass 2 kg and length 1 m pivoted at its end C on a vertical wall as shown in figure. It is supported by a cable AB such that the system is in equilibrium. The tension in the cable is (Take g = 10 m/s^2)

Open solution



(A) 90 N (B) 240 N (C) 30 N (D) 300 N

Q46. JEE Main 2022 (Online) 29th July Evening Shift

The torque of a force $5\hat{i}+3\hat{j}-7\hat{k}$ about the origin is τ. If the force acts on a particle whose position vector is $2\hat{i}+2\hat{j}+\hat{k}$, then the value of τ will be

(A) $11\hat{i}+19\hat{j}-4\hat{k}$ (B) $-11\hat{i}+9\hat{j}-16\hat{k}$ (C) $-17\hat{i}+19\hat{j}-4\hat{k}$ (D) $17\hat{i}+9\hat{j}+16\hat{k}$

Open solution

Q47. JEE Main 2022 (Online) 25th July Morning Shift

A solid cylinder and a solid sphere, having same mass M and radius R, roll down the same inclined plane from top without slipping. They start from rest. The ratio of velocity of the solid cylinder to that of the solid sphere, with which they reach the ground, will be :

(A) $\sqrt{\frac{5}{3}}$ (B) $\sqrt{\frac{4}{5}}$ (C) $\sqrt{\frac{3}{5}}$ (D) $\sqrt{\frac{14}{15}}$

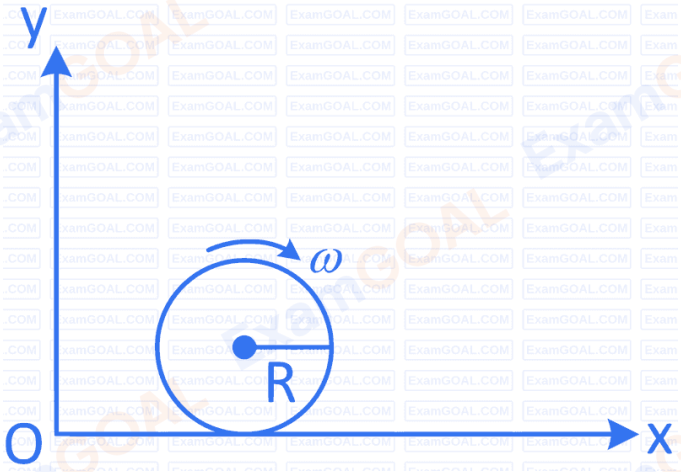
Open solution

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Q48. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3}$ $R^2\omega$. The value of a will be :



(A) 2 (B) 3 (C) 5 (D) 4

Q49. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

A ball is spun with angular acceleration α = $6t^2 - 2t$ where t is in second and α is in $rads^{-2}$. At t = 0, the ball has angular velocity of 10 $rads^{-1}$ and angular position of 4 rad. The most appropriate expression for the angular position of the ball is :

(A) $\frac{3}{2}t^4 - t^2 + 10t$

(B) $\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

(C) $\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$

(D) $2t^4 - \frac{t^3}{2} + 5t + 4$

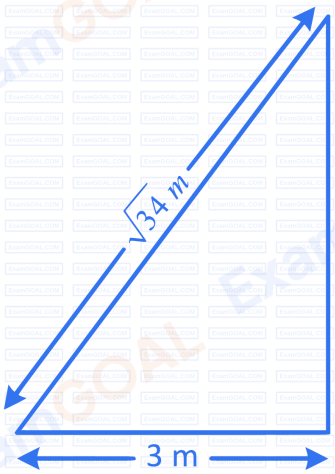
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Q50. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

A $\sqrt{34}$ m long ladder weighing 10 kg leans on a frictionless wall. Its feet rest on the floor 3 m away from the wall as shown in the figure. If F_f and F_w are the reaction forces of the floor and the wall, then ratio of F_w/F_f will be :

(Use g = 10 m/s^2.)



(A) $\frac{6}{\sqrt{110}}$ (B) $\frac{3}{\sqrt{113}}$ (C) $\frac{3}{\sqrt{109}}$ (D) $\frac{2}{\sqrt{109}}$

Q51. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Match List-I with List-II

	List-I		List-II
(A)	Moment of inertia of solid sphere of radius R about any tangent.	(I)	$\frac{5}{3}MR^2$
(B)	Moment of inertia of hollow sphere of radius (R) about any tangent.	(II)	$\frac{7}{5}MR^2$
(C)	Moment of inertia of circular ring of radius (R) about its diameter.	(III)	$\frac{1}{4}MR^2$
(D)	Moment of inertia of circular disc of radius (R) about any diameter.	(IV)	$\frac{1}{2}MR^2$

Choose the correct answer from the options given below :

(A) A - II, B - I, C - IV, D - III

(B) A - I, B - II, C - IV, D - III

(C) A - II, B - I, C - III, D - IV

(D) A - I, B - II, C - III, D - IV

Q52. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

One end of a massless spring of spring constant k and natural length l_0 is fixed while the other end is connected to a small object of mass m lying on a frictionless table. The spring remains horizontal on the table. If the object is made to rotate at an angular velocity ω about an axis passing through fixed end, then the elongation of the spring will be :

(A) $\frac{k-m\omega^2 l_0}{m\omega^2}$ (B) $\frac{m\omega^2 l_0}{k+m\omega^2}$

(C) $\frac{m\omega^2 l_0}{k-m\omega^2}$ (D) $\frac{k+m\omega^2 l_0}{m\omega^2}$

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Q53. JEE Main 2022 (Online) 26th June Evening Shift

A solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is

(A) $\frac{2}{5}$ (B) $\frac{2}{7}$ (C) $\frac{1}{5}$ (D) $\frac{7}{10}$

Open solution

Q54. JEE Main 2022 (Online) 26th June Morning Shift

A thin circular ring of mass M and radius R is rotating with a constant angular velocity 2 rads^{-1} in a horizontal plane about an axis vertical to its plane and passing through the center of the ring. If two objects each of mass m be attached gently to the opposite ends of a diameter of ring, the ring will then rotate with an angular velocity (in rads^{-1}).

(A) $\frac{M}{(M+m)}$ (B) $\frac{(M+2m)}{2M}$ (C) $\frac{2M}{(M+2m)}$ (D) $\frac{2(M+2m)}{M}$

Open solution

Q55. JEE Main 2022 (Online) 25th June Morning Shift

If force $\overrightarrow{F} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ acts on a particle position vector $2\hat{i} + \hat{j} + 2\hat{k}$ then, the torque about the origin will be :

(A) $3\hat{i} + 4\hat{j} - 2\hat{k}$ (B) $-10\hat{i} + 10\hat{j} + 5\hat{k}$

(C) $10\hat{i} + 5\hat{j} - 10\hat{k}$ (D) $10\hat{i} + \hat{j} - 5\hat{k}$

Open solution

Q56. JEE Main 2022 (Online) 24th June Evening Shift

A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be :

(A) 7.5 rad (B) 15 rad (C) 20 rad (D) 30 rad

Open solution

Q57. JEE Main 2021 (Online) 31st August Evening Shift

A system consists of two identical spheres each of mass 1.5 kg and radius 50 cm at the ends of a light rod. The distance between the centres of the two spheres is 5 m. What will be the moment of inertia of the system about an axis perpendicular to the rod passing through its midpoint?

(A) 18.75 kgm^2 (B) 1.905×10^5 kgm^2

(C) 19.05 kgm^2 (D) 1.875×10^5 kgm^2

Open solution

Q58. JEE Main 2021 (Online) 31st August Morning Shift

Angular momentum of a single particle moving with constant speed along circular path :

(A) changes in magnitude but remains same in the direction

(B) remains same in magnitude and direction

(C) remains same in magnitude but changes in the direction

(D) is zero

Open solution

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Q59. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

Two discs have moments of inertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by :

(A) $\frac{I_1 I_2}{(I_1+I_2)}(\omega_1-\omega_2)^2$

(B) $\frac{(I_1-I_2)^2\omega_1\omega_2}{2(I_1+I_2)}$

(C) $\frac{I_1 I_2}{2(I_1+I_2)}(\omega_1-\omega_2)^2$

(D) $\frac{(\omega_1-\omega_2)^2}{2(I_1+I_2)}$

Q60. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

Moment of inertia of a square plate of side l about the axis passing through one of the corner and perpendicular to the plane of square plate is given by :

(A) $\frac{Ml^2}{6}$

(B) Ml^2

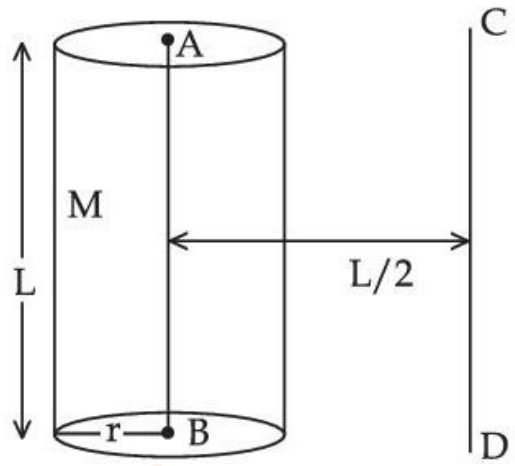
(C) $\frac{Ml^2}{12}$

(D) $\frac{2}{3}Ml^2$

Q61. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

The solid cylinder of length 80 cm and mass M has a radius of 20 cm. Calculate the density of the material used if the moment of inertia of the cylinder about an axis CD parallel to AB as shown in figure is 2.7 kg m^2.



(A) 14.9 kg/m^3

(B) 7.5×10^1 kg/m^3

(C) 7.5×10^2 kg/m^3

(D) 1.49×10^2 kg/m^3

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Q62. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

List-I	List-II
(a) MI of the rod (length L, Mass M, about an axis $\perp$ to the rod passing through the midpoint)	(i) $8ML^2/3$
(b) MI of the rod (length L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(ii) $ML^2/3$
(c) MI of the rod (length 2L, Mass M, about an axis $\perp$ to the rod passing through its midpoint)	(iii) $ML^2/12$
(d) MI of the rod (Length 2L, Mass 2M, about an axis $\perp$ to the rod passing through one of its end)	(iv) $2ML^2/3$

Choose the correct answer from the options given below:

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

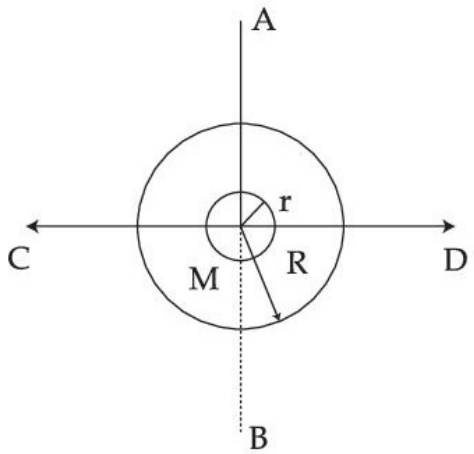
(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q63. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

The figure shows two solid discs with radius R and r respectively. If mass per unit area is same for both, what is the ratio of MI of bigger disc around axis AB (Which is $\perp$ to the plane of the disc and passing through its centre) to MI of smaller disc around one of its diameters lying on its plane? Given 'M' is the mass of the larger disc. (MI stands for moment of inertia)



(A) $R^2 : r^2$

(B) $2r^4 : R^4$

(C) $2R^2 : r^2$

(D) $2R^4 : r^4$

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Q64. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Moment of inertia of a circular disc of mass 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x, I_y and I_z respectively. The respectively radii of gyration about all the three axes will be the same.

Reason R : A rigid body making rotational motion has fixed mass and shape. In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct but R is NOT the correct explanation of A.

(B) A is not correct but R is correct.

(C) A is correct but R is not correct.

(D) Both A and R are correct and R is the correct explanation of A.

Q65. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Consider a situation in which a ring, a solid cylinder and a solid sphere roll down on the same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter.

The correct statement for this situation is

(A) All of them will have same velocity.

(B) The ring has greatest and the cylinder has the least velocity of the centre of mass at the bottom of the inclined plane.

(C) The sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane.

(D) The cylinder has the greatest and the sphere has the least velocity of the centre of mass at the bottom of the inclined plane.

Q66. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

A body rolls down an inclined plane without slipping. The kinetic energy of rotation is 50% of its translational kinetic energy. The body is :

(A) Solid sphere (B) Solid cylinder (C) Hollow cylinder (D) Ring

Q67. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Consider a uniform wire of mass M and length L. It is bent into a semicircle. Its moment of inertia about a line perpendicular to the plane of the wire passing through the center is :

(A) $\frac{1}{4}\frac{ML^2}{\pi^2}$ (B) $\frac{1}{2}\frac{ML^2}{\pi^2}$

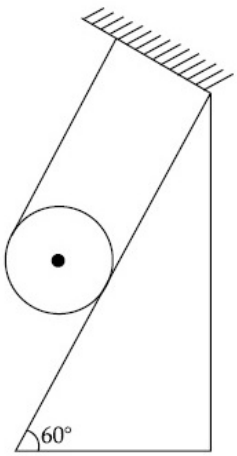
(C) $\frac{2}{5}\frac{ML^2}{\pi^2}$ (D) $\frac{ML^2}{\pi^2}$

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Q68. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

A solid cylinder of mass m is wrapped with an inextensible light string and, is placed on a rough inclined plane as shown in the figure. The frictional force acting between the cylinder and the inclined plane is :



[The coefficient of static friction, μ_s is 0.4]

(A) 0 (B) 5 mg (C) $\frac{7}{2}$ mg (D) $\frac{mg}{5}$

Q69. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

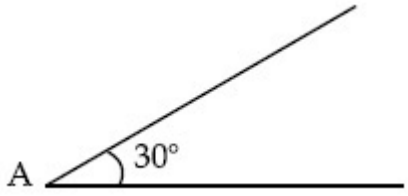
A thin circular ring of mass M and radius r is rotating about its axis with an angular speed ω. Two particles having mass m each are now attached at diametrically opposite points. The angular speed of the ring will become :

(A) $\omega\frac{M}{M+m}$ (B) $\omega\frac{M+2m}{M}$ (C) $\omega\frac{M}{M+2m}$ (D) $\omega\frac{M-2m}{M+2m}$

Q70. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

A sphere of mass 2 kg and radius 0.5 m is rolling with an initial speed of 1 ms^{-1} goes up an inclined plane which makes an angle of 30° with the horizontal plane, without slipping. How long will the sphere take to return to the starting point A?



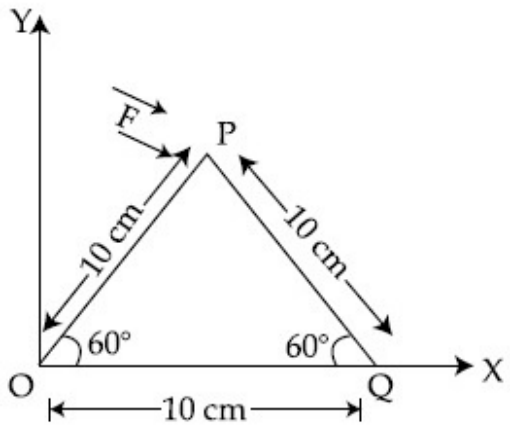
(A) 0.60 s (B) 0.52 s (C) 0.80 s (D) 0.57 s

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Q71. JEE Main 2021 (Online) 17th March Morning Shift

A triangular plate is shown. A force $\vec{F} = 4\hat{i} - 3\hat{j}$ is applied at point P. The torque at point P with respect to point 'O' and 'Q' are :

Open solution



(A) $-15 + 20\sqrt{3}, 15 + 20\sqrt{3}$

(B) $15 - 20\sqrt{3}, 15 + 20\sqrt{3}$

(C) $15 + 20\sqrt{3}, 15 - 20\sqrt{3}$

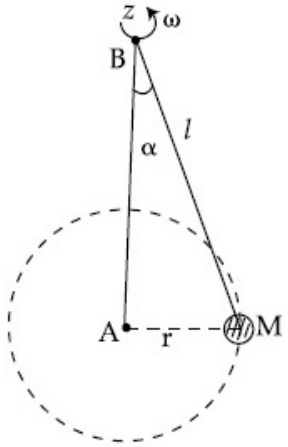
(D) $-15 - 20\sqrt{3}, 15 - 20\sqrt{3}$

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Q72. JEE Main 2021 (Online) 17th March Morning Shift

Open solution

A mass M hangs on a massless rod of length l which rotates at a constant angular frequency. The mass M moves with steady speed in a circular path of constant radius. Assume that the system is in steady circular motion with constant angular velocity ω. The angular momentum of M about point A is L_A which lies in the positive z direction and the angular momentum of M about point B is L_B. The correct statement for this system is :



(A) L_A is constant, both in magnitude and direction

(B) L_B is constant in direction with varying magnitude

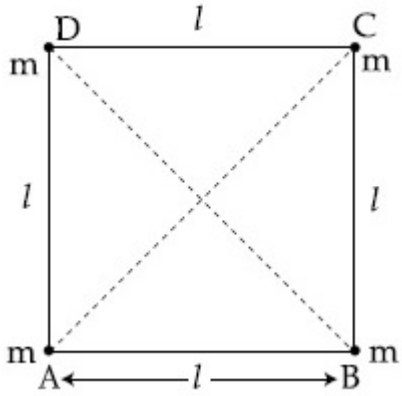
(C) L_B is constant, both in magnitude and direction

(D) L_A and L_B are both constant in magnitude and direction

Q73. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

Four equal masses, m each are placed at the corners of a square of length (l) as shown in the figure. The moment of inertia of the system about an axis passing through A and parallel to DB would be :



(A) $\sqrt{3}\,ml^2$ (B) $2\,ml^2$ (C) ml^2 (D) $3\,ml^2$

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Q74. JEE Main 2021 (Online) 26th February Evening Shift

A cord is wound round the circumference of wheel of radius r. The axis of the wheel is horizontal and the moment of inertia about it is I. A weight mg is attached to the cord at the end. The weight falls from rest. After falling through a distance 'h', the square of angular velocity of wheel will be :

Open solution

(A) $\frac{2mgh}{I+2mr^2}$ (B) $\frac{2mgh}{I+mr^2}$ (C) 2gh (D) $\frac{2gh}{I+mr^2}$

Q75. JEE Main 2021 (Online) 26th February Morning Shift

Four identical solid spheres each of mass 'm' and radius 'a' are placed with their centres on the four corners of a square of side 'b'. The moment of inertia of the system about one side of square where the axis of rotation is parallel to the plane of the square is :

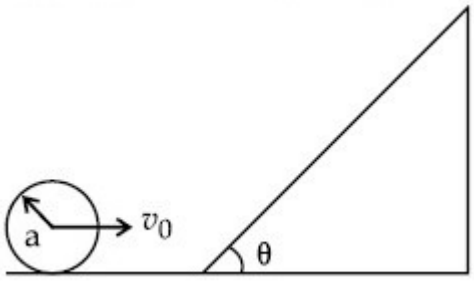
Open solution

(A) $\frac{4}{5}ma^2$ (B) $\frac{8}{5}ma^2+mb^2$ (C) $\frac{4}{5}ma^2+2mb^2$ (D) $\frac{8}{5}ma^2+2mb^2$

Q76. JEE Main 2021 (Online) 25th February Evening Shift

A sphere of radius 'a' and mass 'm' rolls along a horizontal plane with constant speed v_0. It encounters an inclined plane at angle θ and climbs upward. Assuming that it rolls without slipping, how far up the sphere will travel?

Open solution



(A) $\frac{v_0^2}{2g\sin\theta}$ (B) $\frac{7v_0^2}{10g\sin\theta}$

(C) $\frac{2}{5}\frac{v_0^2}{g\sin\theta}$ (D) $\frac{v_0^2}{5g\sin\theta}$

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Q77. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

Moment of inertia (M. I.) of four bodies, having same mass and radius, are reported as;

I_1 = M.I. of thin circular ring about its diameter,

I_2 = M.I. of circular disc about an axis perpendicular to disc and going through the centre,

I_3 = M.I. of solid cylinder about its axis and

I_4 = M.I. of solid sphere about its diameter.

Then :

(A) $I_1 = I_2 = I_3 > I_4$

(B) $I_1 + I_3 < I_2 + I_4$

(C) $I_1 = I_2 = I_3 < I_4$

(D) $I_1 + I_2 = I_3 + \frac{5}{2} I_4$

Q78. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

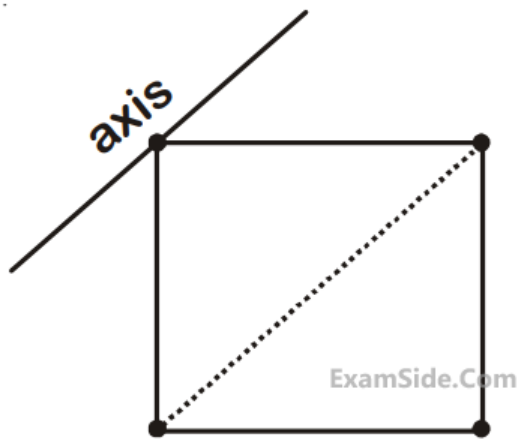
The linear mass density of a thin rod AB of length L varies from A to B as $\lambda(x) = \lambda_0\left(1 + \frac{x}{L}\right)$, where x is the distance from A. If M is the mass of the rod then its moment of inertia about an axis passing through A and perpendicular to the rod is :

(A) $\frac{2}{5}ML^2$ (B) $\frac{5}{12}ML^2$ (C) $\frac{7}{18}ML^2$ (D) $\frac{3}{7}ML^2$

Q79. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

Four point masses, each of mass m, are fixed at the corners of a square of side l. The square is rotating with angular frequency ω, about an axis passing through one of the corners of the square and parallel to its diagonal, as shown in the figure. The angular momentum of the square about this axis is :



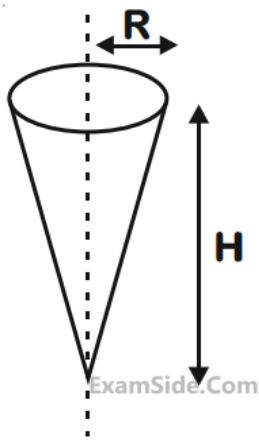
(A) $3ml^2\omega$ (B) $4ml^2\omega$ (C) $ml^2\omega$ (D) $2ml^2\omega$

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Q80. JEE Main 2020 (Online) 6th September Morning Slot

Shown in the figure is a hollow icecream cone (it is open at the top). If its mass is M, radius of its top, R and height, H, then its moment of inertia about its axis is :

Open solution



(A) $\frac{M(R^2+H^2)}{3}$ (B) $\frac{MR^2}{2}$
(C) $\frac{MR^2}{3}$ (D) $\frac{MH^2}{3}$

Q81. JEE Main 2020 (Online) 5th September Evening Slot

A ring is hung on a nail. It can oscillate, without slipping or sliding
(i) in its plane with a time period T_1 and,
(ii) back and forth in a direction perpendicular to its plane,
with a period T_2. The ratio $\frac{T_1}{T_2}$ will be :

Open solution

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{2}{3}$ (D) $\frac{3}{\sqrt{2}}$

Q82. JEE Main 2020 (Online) 5th September Morning Slot

A wheel is rotating freely with an angular speed ω on a shaft. The moment of inertia of the wheel is I and the moment of inertia of the shaft is negligible. Another wheel of moment of inertia 3I initially at rest is suddenly coupled to the same shaft. The resultant fractional loss in the kinetic energy of the system is :

Open solution

(A) 0 (B) $\frac{5}{6}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

Q83. JEE Main 2020 (Online) 4th September Evening Slot

Consider two uniform discs of the same thickness and different radii R_1 = R and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of inertia I_1 and I_2, respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is :

Open solution

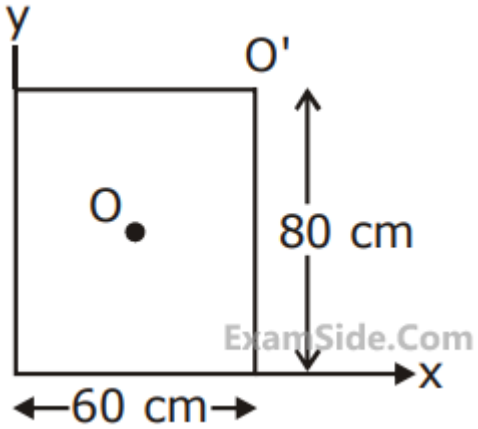
(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4

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Q84. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

For a uniform rectangular sheet shown in the figure, the ratio of moments of inertia about the axes perpendicular to the sheet and passing through O (the centre of mass) and O' (corner point) is :

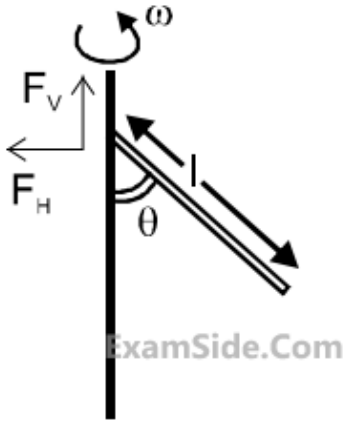


(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{2}{3}$

Q85. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

A uniform rod of length 'l' is pivoted at one of its ends on a vertical shaft of negligible radius. When the shaft rotates at angular speed ω the rod makes an angle θ with it (see figure). To find θ equate the rate of change of angular momentum (direction going into the paper) $\frac{ml^2}{12}\omega^2 \sin\theta\cos\theta$ about the centre of mass (CM) to the torque provided by the horizontal and vertical forces F_H and F_V about the CM. The value of θ is then such that :



(A) $\cos\theta = \frac{2g}{3l\omega^2}$ (B) $\cos\theta = \frac{3g}{2l\omega^2}$

(C) $\cos\theta = \frac{g}{2l\omega^2}$ (D) $\cos\theta = \frac{g}{l\omega^2}$

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Q86. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

Moment of inertia of a cylinder of mass M, length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is
I = $M\left(\frac{R^2}{4}+\frac{L^2}{12}\right)$. If such a cylinder is to be made for a given mass of a material, the ratio $\frac{L}{R}$ for it to have minimum possible I is

(A) $\frac{3}{2}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{\frac{2}{3}}$ (D) $\frac{2}{3}$

Q87. JEE Main 2020 (Online) 2nd September Evening Slot

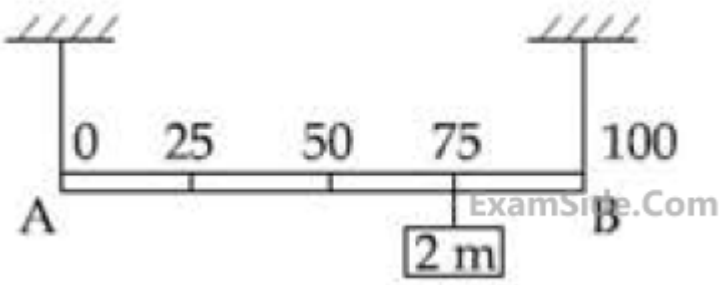
Open solution

Two uniform circular discs are rotating independently in the same direction around their common axis passing through their centres. The moment of inertia and angular velocity of the first disc are 0.1 kg-m^2 and 10 rad s^{-1} respectively while those for the second one are 0.2 kg-m^2 and 5 rad s^{-1} respectively. At some instant they get stuck together and start rotating as a single system about their common axis with some angular speed. The kinetic energy of the combined system is :

(A) $\frac{20}{3}J$ (B) $\frac{5}{3}J$ (C) $\frac{10}{3}J$ (D) $\frac{2}{3}J$

Q88. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution



Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. The tension in the string at A is :

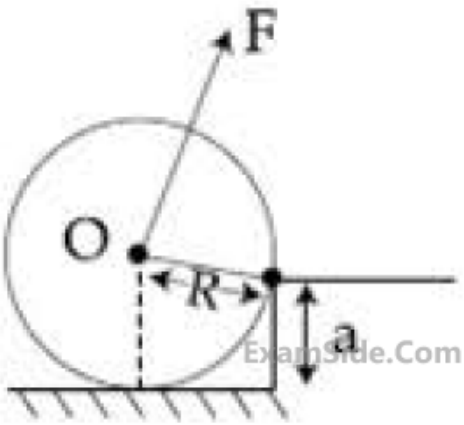
(A) 0.5 mg (B) 2 mg (C) 0.75 mg (D) 1 mg

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Q89. JEE Main 2020 (Online) 2nd September Morning Slot

A uniform cylinder of mass M and radius R is to be pulled over a step of height a (a < R) by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is :

Open solution



(A) $Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$

(B) $Mg\sqrt{1-\frac{a^2}{R^2}}$

(C) $Mg\frac{a}{R}$

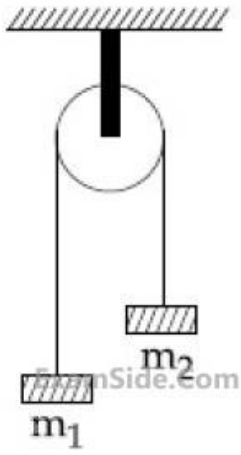
(D) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2-1}$

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Q90. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

A uniformly thick wheel with moment of inertia I and radius R is free to rotate about its centre of mass (see fig). A massless string is wrapped over its rim and two blocks of masses m_1 and m_2 ($m_1 > m_2$) are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when m_1 descents by a distance h is :



(A) $\left[\frac{2(m_1+m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$

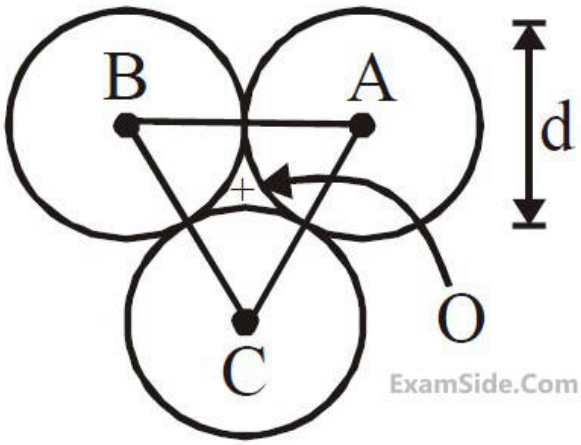
(B) $\left[\frac{m_1+m_2}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}} gh$

(C) $\left[\frac{(m_1-m_2)}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}} gh$

(D) $\left[\frac{2(m_1-m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$

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Q91. JEE Main 2020 (Online) 9th January Morning Slot



Open solution

Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of side of length d. The ratio I_0/I_A of moment of inertia I_0 of the system about an axis passing through the centroid and about center of any of the spheres I_A and perpendicular to the plane of the triangle is :

(A) $\frac{13}{23}$ (B) $\frac{23}{13}$ (C) $\frac{15}{13}$ (D) $\frac{13}{15}$

Q92. JEE Main 2020 (Online) 8th January Evening Slot

A uniform sphere of mass 500 g rolls without slipping on a plane horizontal surface with its centre moving at a speed of 5.00 cm/s. Its kinetic energy is :

(A) 8.75×10^{-3} J (B) 1.13×10^{-3} J (C) 8.75×10^{-4} J (D) 6.25×10^{-4} J

Open solution

Q93. JEE Main 2020 (Online) 8th January Morning Slot

Consider a uniform rod of mass M = 4m and length ℓ pivoted about its centre. A mass m moving with velocity v making angle $\theta = \frac{\pi}{4}$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod-mass system just after the collision is :

(A) $\frac{3\sqrt{2}}{7}\frac{v}{\ell}$ (B) $\frac{3}{7}\frac{v}{\ell}$

(C) $\frac{3}{7\sqrt{2}}\frac{v}{\ell}$ (D) $\frac{4}{7}\frac{v}{\ell}$

Open solution

Q94. JEE Main 2020 (Online) 7th January Evening Slot

Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r)$ = A + Br . The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is:

(A) $2\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$ (B) $\pi a^4\left(\frac{A}{4}+\frac{aB}{5}\right)$

(C) $2\pi a^4\left(\frac{aA}{4}+\frac{B}{5}\right)$ (D) $2\pi a^4\left(\frac{A}{4}+\frac{B}{5}\right)$

Open solution

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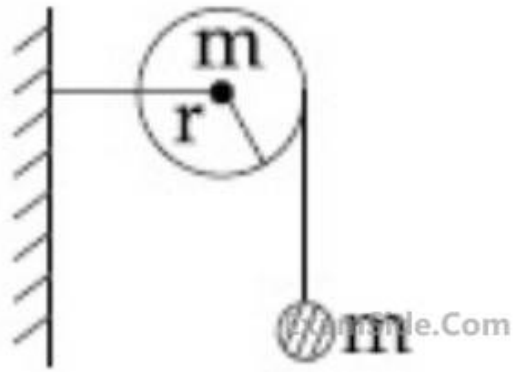
Q95. JEE Main 2020 (Online) 7th January Morning Slot

The radius of gyration of a uniform rod of length l, about an axis passing through a point $\frac{l}{4}$ away from the centre of the rod, and perpendicular to it, is :

(A) $\frac{1}{8}l$ (B) $\frac{1}{4}l$ (C) $\sqrt{\frac{7}{48}}l$ (D) $\sqrt{\frac{3}{8}}l$

Open solution

Q96. JEE Main 2020 (Online) 7th January Morning Slot



Open solution

As shown in the figure, a bob of mass m is tied by a massless string whose other end portion is wound on a fly wheel (disc) of radius r and mass m. When released from rest the bob starts falling vertically. When it has covered a distance of h, the angular speed of the wheel will be:

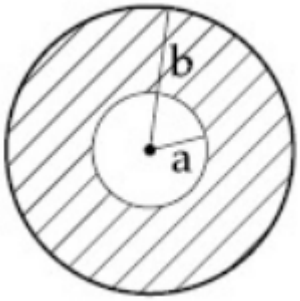
(A) $r\sqrt{\frac{3}{2gh}}$ (B) $r\sqrt{\frac{3}{4gh}}$

(C) $\frac{1}{r}\sqrt{\frac{4gh}{3}}$ (D) $\frac{1}{r}\sqrt{\frac{2gh}{3}}$

Q97. JEE Main 2019 (Online) 12th April Morning Slot

A circular disc of radius b has a hole of radius a at its centre (see figure). If the mass per unit area of the disc varies as $\left(\frac{\sigma_0}{r}\right)$, then the radius of gyration of the disc about its axis passing through the centre is:

Open solution



(A) $\sqrt{\frac{a^2+b^2+ab}{2}}$ (B) $\sqrt{\frac{a+b}{3}}$

(C) $\sqrt{\frac{a^2+b^2+ab}{3}}$ (D) $\sqrt{\frac{a+b}{2}}$

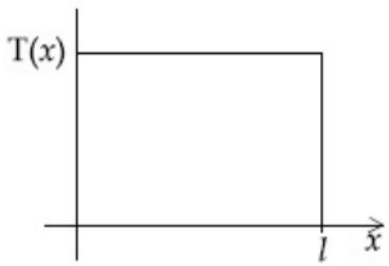
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Q98. JEE Main 2019 (Online) 12th April Morning Slot

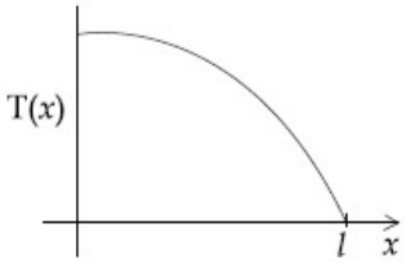
Open solution

A uniform rod of length ℓ is being rotated in a horizontal plane with a constant angular speed about an axis passing through one of its ends. If the tension generated in the rod due to rotation is T(x) at a distance x from the axis, then which of the following graphs depicts it most closely?

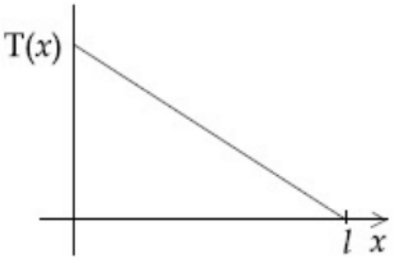
(A)



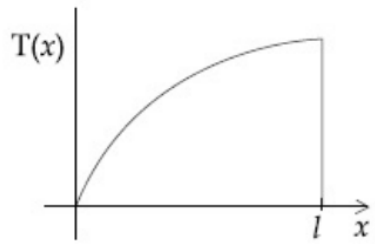
(B)



(C)



(D)



Q99. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

A person of mass M is, sitting on a swing of length L and swinging with an angular amplitude θ_0. If the person stands up when the swing passes through its lowest point, the work done by him, assuming that his center of mass moves by a distance $\ell(\ell << L)$, is close to;

(A) $mg\ell(1 + \theta_0^2)$

(B) $mg\ell$

(C) $mg\ell(1 + \frac{\theta_0^2}{2})$

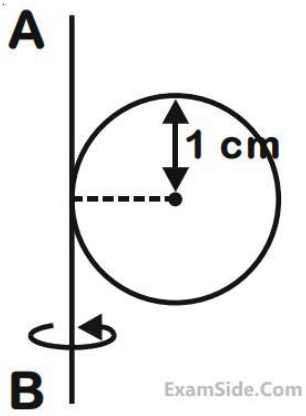
(D) $mg\ell(1 - \theta_0^2)$

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Q100. JEE Main 2019 (Online) 10th April Evening Slot

A metal coin of mass 5 g and radius 1 cm is fixed to a thin stick AB of negligible mass as shown in the figure. The system is initially at rest. The constant torque, that will make the system rotate about AB at 25 rotations per second in 5s, is close to :

Open solution



(A) 7.9×10^{-6} Nm (B) 4.0×10^{-6} Nm (C) 2.0×10^{-5} Nm (D) 1.6×10^{-5} Nm

Q101. JEE Main 2019 (Online) 10th April Evening Slot

A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is converted into a uniform disc of radius 2R. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the disc about its axis and I_2 be the moment of inertia of the new sphere about its axis. The ratio I_1/I_2 is given by :

Open solution

(A) 65 (B) 140 (C) 185 (D) 285

Q102. JEE Main 2019 (Online) 10th April Evening Slot

The time dependence of the position of a particle of mass m = 2 is given by $\vec{r}(t) = 2t\hat{i} - 3t^2\hat{j}$. Its angular momentum, with respect to the origin, at time t = 2 is

Open solution

(A) $36\,\hat{k}$ (B) $-48\,\hat{k}$
(C) $-34\left(\hat{k}-\hat{i}\right)$ (D) $48\left(\hat{i}+\hat{j}\right)$

Q103. JEE Main 2019 (Online) 10th April Morning Slot

A thin disc of mass M and radius R has mass per unit area $\sigma(r) = kr^2$ where r is the distance from its centre. Its moment of inertia about an axis going through its centre of mass and perpendicular to its plane is :

Open solution

(A) $\frac{MR^2}{3}$ (B) $\frac{MR^2}{6}$ (C) $\frac{2MR^2}{3}$ (D) $\frac{MR^2}{2}$

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Q104. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

Two coaxial discs, having moments of inertia I_1 and $I_1/2$, are rotating with respective angular velocities ω_1 and $\omega_1/2$, about their common axis. They are brought in contact with each other and thereafter they rotate with a common angular velocity. If E_f and E_i are the final and initial total energies, then $(E_f - E_i)$ is:

(A) $\frac{I_1\omega_1^2}{24}$ (B) $\frac{I_1\omega_1^2}{12}$ (C) $\frac{3}{8}I_1\omega_1^2$ (D) $\frac{I_1\omega_1^2}{6}$

Q105. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

A particle of mass m is moving along a trajectory given by

$x = x_0 + a\cos\omega_1 t$

$y = y_0 + b\sin\omega_2 t$

The torque, acting on the particle about the origin, at t = 0 is :

(A) Zero (B) $+my_0a\,\omega_1^2\hat{k}$

(C) $-m\left(x_0b\omega_2^2 - y_0a\omega_1^2\right)\hat{k}$ (D) $m\,(-x_0b + y_0a)\,\omega_1^2\hat{k}$

Q106. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Moment of inertia of a body about a given axis is 1.5 kg m^2. Initially the body is at rest. In order to produce a rotational kinetic energy of 1200 J, the angular accleration of 20 rad/s^2 must be applied about the axis for a duration of :-

(A) 2.5 s (B) 3 s (C) 5s (D) 2 s

Q107. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

A thin smooth rod of length L and mass M is rotating freely with angular speed ω_0 about an axis perpendicular to the rod and passing through its center. Two beads of mass m and negligible size are at the center of the rod initially. The beads are free to slide along the rod. The angular speed of the system , when the beads reach the opposite ends of the rod, will be :-

(A) $\frac{M\omega_0}{M+3m}$ (B) $\frac{M\omega_0}{M+m}$ (C) $\frac{M\omega_0}{M+6m}$ (D) $\frac{M\omega_0}{M+2m}$

Q108. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

The following bodies are made to roll up (without slipping) the same inclined plane from a horizontal plane. : (i) a ring of radius R, (ii) a solid cylinder of radius R/2 and (iii) a solid sphere of radius R/4 . If in each case, the speed of the centre of mass at the bottom of the incline is same, the ratio of the maximum heights they climb is :

(A) 20 : 15 : 14 (B) 4 : 3 : 2 (C) 2 : 3 : 4 (D) 10 : 15 : 7

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Q109. JEE Main 2019 (Online) 9th April Morning Slot

A stationary horizontal disc is free to rotate about its axis. When a torque is applied on it, its kinetic energy as a function of θ, where θ is the angle by which it has rotated, is given as $k\theta^2$. If its moment of inertia is I then the angular acceleration of the disc is :

Open solution

(A) $\frac{k}{4I}\theta$ (B) $\frac{k}{I}\theta$ (C) $\frac{k}{2I}\theta$ (D) $\frac{2k}{I}\theta$

Q110. JEE Main 2019 (Online) 8th April Evening Slot

A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights h_{sph} and h_{cyl} on the incline. The ratio h_{sph}/h_{cyl} is given by :-

Open solution



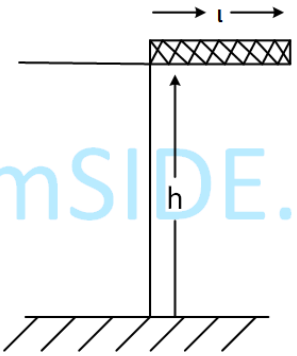
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(A) 1 (B) 14/15 (C) 4/5 (D) $2/\sqrt{5}$

Q111. JEE Main 2019 (Online) 8th April Evening Slot

A rectangular solid box of length 0.3 m is held horizontally, with one of its sides on the edge of a platform of height 5m. When released, it slips off the table in a very short time t = 0.01s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to :-

Open solution



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(A) 0.28 (B) 0.02 (C) 0.3 (D) 0.5

Q112. JEE Main 2019 (Online) 8th April Morning Slot

A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its centre. The moment of Inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = aMR^2$. The value of the coefficient a is :

Open solution

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{8}{5}$ (D) $\frac{3}{5}$

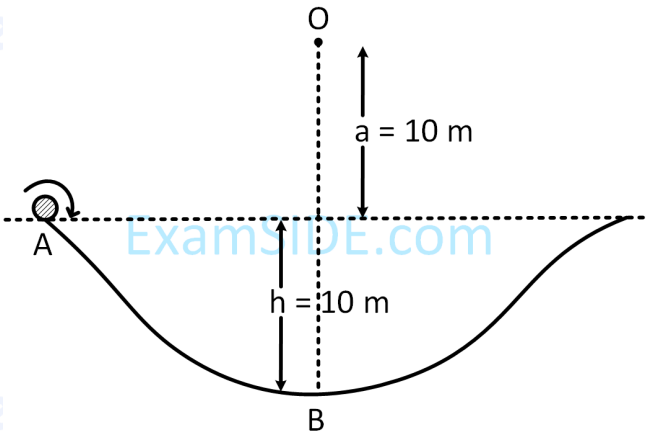
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Q113. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

A particle of mass 20 g is released with an initial velocity 5 m/s along the curve from the point A, as shown in the figure. The point A is at height h from point B. The particle slides along the frictionless surface. When the particle reaches point B, its angular momentum about O will be :

(Take g = 10 m/s^2)



(A) 6 kg-m^2/s (B) 8 kg-m^2/s (C) 2 kg-m^2/s (D) 3 kg-m^2/s

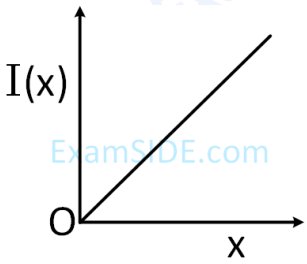
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Q114. JEE Main 2019 (Online) 12th January Evening Slot

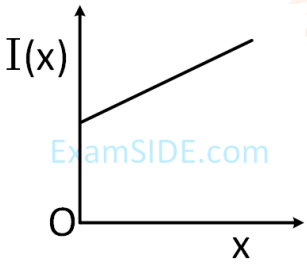
Open solution

The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is 'I(x)'. Which one of the graphs represents the variation of I(x) with x correctly ?

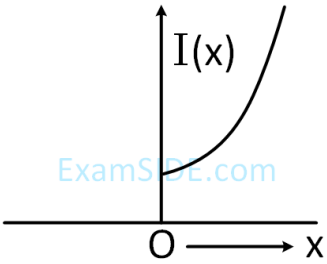
(A)



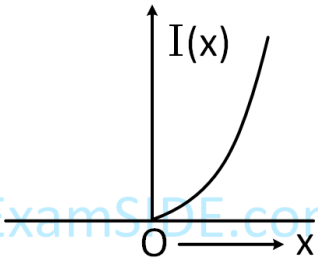
(B)



(C)



(D)

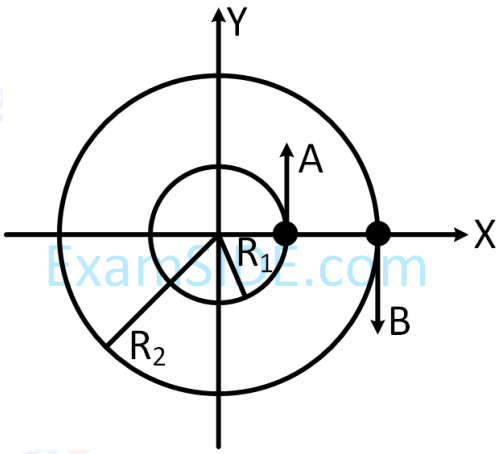


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Q115. JEE Main 2019 (Online) 12th January Evening Slot

Two particles A, B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω. At t = 0, their positions and direction of motion are shown in the figure. :

Open solution



The relative velocity $\overrightarrow{V_A} - \overrightarrow{V_B}$ at t = $\frac{\pi}{2\omega}$ is given by :

(A) $-\omega(R_1 + R_2)\hat{i}$

(B) $\omega(R_2 - R_1)\hat{i}$

(C) $\omega(R_1 + R_2)\hat{i}$

(D) $\omega(R_1 - R_2)\hat{i}$

Q116. JEE Main 2019 (Online) 12th January Morning Slot

Let the moment of inertia of a hollow cylinder of length 30 cm (inner radius 10 cm and outer radius 20 cm), about its axis be I. The radius of a thin cylinder of the same mass such that its moment of inertia about its axis is also I, is :

Open solution

(A) 16 cm

(B) 12 cm

(C) 14 cm

(D) 18 cm

Q117. JEE Main 2019 (Online) 11th January Evening Slot

The magnitude of torque on a particle of mass 1 kg is 2.5 Nm about the origin. If the force acting on it is 1 N, and the distance of the particle from the origin is 5m, the angle between the force and the position vector is (in radians) :

Open solution

(A) $\frac{\pi}{8}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{4}$

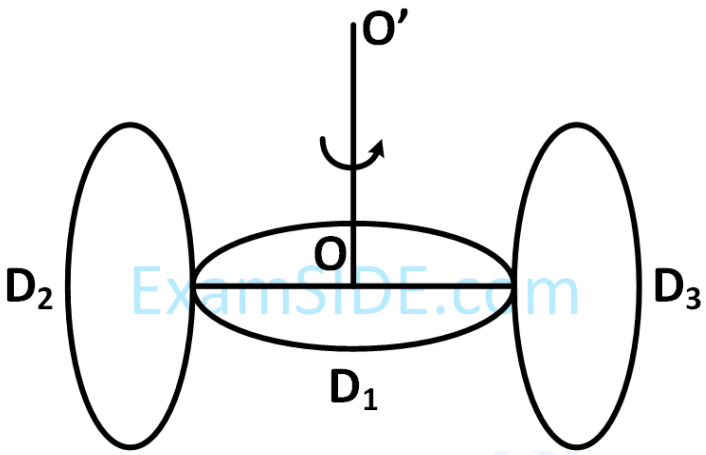
(D) $\frac{\pi}{3}$

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Q118. JEE Main 2019 (Online) 11th January Evening Slot

A circular disc D_1 of mass M and radius R has two identical discs D_2 and D_3 of the same mass M and radius R attached rigidly at its opposite ends (see figure). The moment of inertia of the system about the axis OO' ,passing through the centre of D_1 as shown in the figure, will be :

Open solution

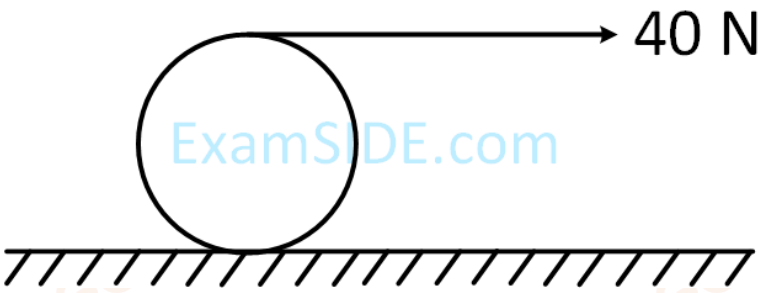


(A) $3MR^2$ (B) MR^2 (C) $\frac{2}{3}MR^2$ (D) $\frac{4}{5}MR^2$

Q119. JEE Main 2019 (Online) 11th January Evening Slot

A string is wound around a hollow cylinder of mass 5 kg and radius 0.5m. If the string is now pulled with a horizontal force of 40 N, and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be (Neglect the mass and thickness of the string) :

Open solution



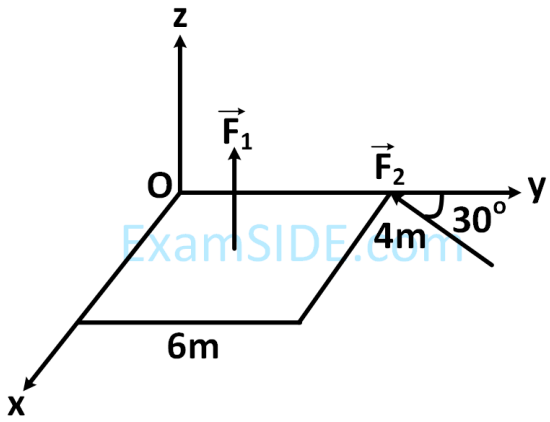
(A) 16 rad/s^2 (B) 20 rad/s^2 (C) 10 rad/s^2 (D) 12 rad/s^2

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Q120. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

A slab is subjected to two forces $\vec{F_1}$ and $\vec{F_2}$ of same magnitude F as shown in the figure. Force $\vec{F_2}$ is in XY-plane while force $\vec{F_1}$ acts along z = axis at the point $\left(2\vec{i}+3\vec{j}\right)$.. The moment of these forces about point O will be :



(A) $\left(3\hat{i}-2\hat{j}-3\hat{k}\right)F$

(B) $\left(3\hat{i}+2\hat{j}-3\hat{k}\right)F$

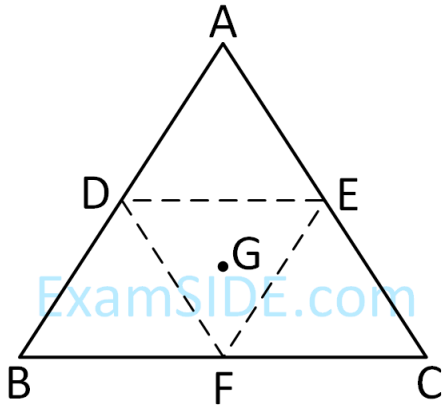
(C) $\left(3\hat{i}+2\hat{j}+3\hat{k}\right)F$

(D) $\left(3\hat{i}-2\hat{j}+3\hat{k}\right)F$

Q121. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

An equilateral triangle ABC is cut from a thin solid sheet of wood. (see figure) D, E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0. If the smaller triangle DEF is removed from ABC, the moment of inertia of the remaining figure about the same axis is I. Then :



(A) $I=\frac{I_0}{4}$

(B) $I=\frac{15}{16}I_0$

(C) $I=\frac{9}{16}I_0$

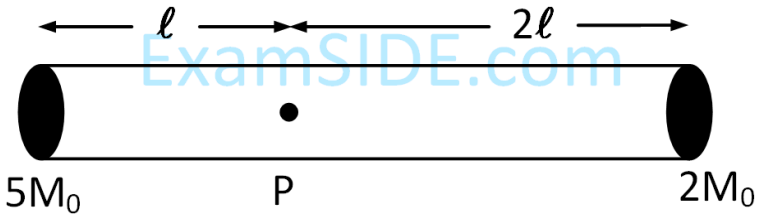
(D) $I=\frac{3}{4}I_0$

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Q122. JEE Main 2019 (Online) 10th January Evening Slot

A rigid massless rod of length 3l has two masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be -

Open solution

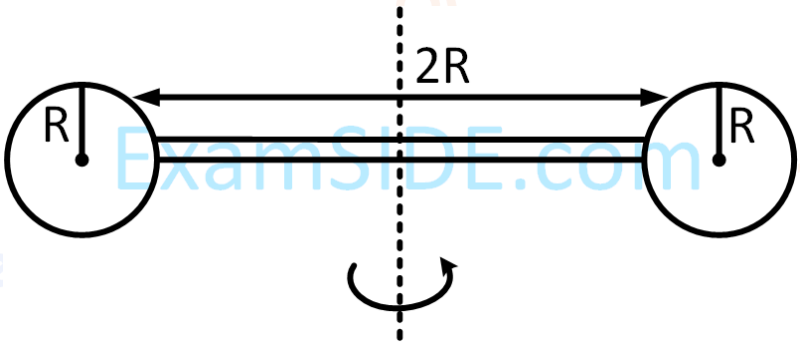


(A) $\frac{g}{13l}$ (B) $\frac{g}{2l}$ (C) $\frac{g}{3l}$ (D) $\frac{7g}{3l}$

Q123. JEE Main 2019 (Online) 10th January Evening Slot

Two identical spherical balls of mass M and radius R each are stuck on two ends of a rod of length 2R and mass M (see figure). The moment of inertia of the system about the axis passing perpendicularly through the centre of the rod is :

Open solution



(A) $\frac{17}{15}$ MR^2 (B) $\frac{137}{15}$ MR^2 (C) $\frac{209}{15}$ MR^2 (D) $\frac{152}{15}$ MR^2

Q124. JEE Main 2019 (Online) 10th January Morning Slot

To mop-clean a floor, a cleaning machine presses a circular mop of radius R vertically down with a total force F and rotates it with a constant angular speed about its axis. If the force F is distributed uniformly over the mop and if coefficient of friction between the mop and the floor is μ, the torque, applied by the machine on the mop is -

Open solution

(A) μFR/2 (B) μFR/3 (C) μFR/6 (D) $\frac{2}{3}\mu$FR

Q125. JEE Main 2019 (Online) 10th January Morning Slot

A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is -

Open solution

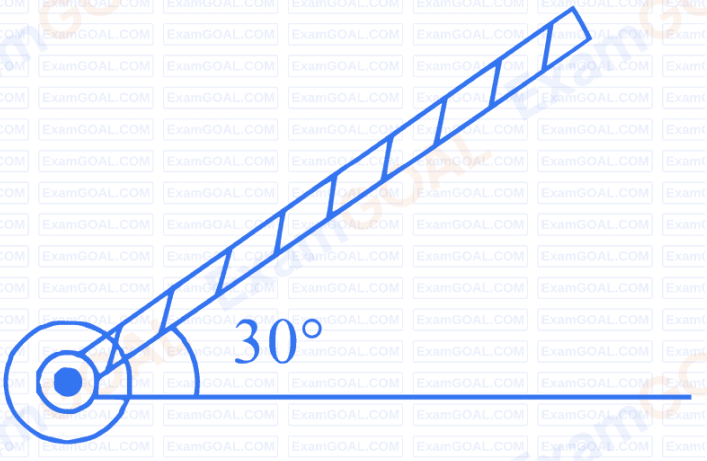
(A) $\frac{F}{2mR}$ (B) $\frac{2F}{3mR}$ (C) $\frac{3F}{2mR}$ (D) $\frac{F}{3mR}$

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Q126. JEE Main 2019 (Online) 9th January Evening Slot

Open solution

A rod of length 50 cm is pivoted at one end. It is raised such that it makes an angle of 30° from the horizontal as shown and released from rest. Its angular speed when it passes through the horizontal (in rad s^{-1}) will be ($g = 10\ ms^{-2}$)

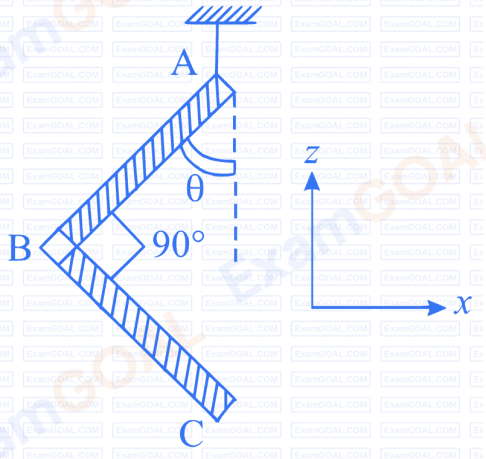


(A) $\sqrt{\frac{30}{7}}$ (B) $\sqrt{30}$ (C) $\frac{\sqrt{20}}{3}$ (D) $\frac{\sqrt{30}}{2}$

Q127. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

An L-shaped object, made of thin rods of uniform mass density, is suspended with a string as shown in figure. If AB = BC, and the angle made by AB with downward vertical is θ, then :



(A) $\tan\theta = \frac{1}{2\sqrt{3}}$ (B) $\tan\theta = \frac{1}{2}$

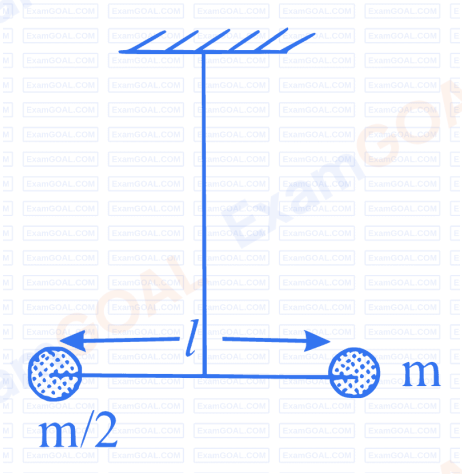
(C) $\tan\theta = \frac{2}{\sqrt{3}}$ (D) $\tan\theta = \frac{1}{3}$

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Q128. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length l. The rod is suspended by a thin wire of torsional constant k, at the centre of mass of the rod-mass system(see figure). Because of torsional constant k, the restoring torque is τ = kθ for angular displacement θ. If the rod is rotated by θ_0 and released, the tension in it when it passes through its mean position will be :



(A) $\frac{3k{\theta_0}^2}{l}$ (B) $\frac{2k{\theta_0}^2}{l}$ (C) $\frac{k{\theta_0}^2}{l}$ (D) $\frac{k{\theta_0}^2}{2l}$

Q129. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

If the angular momentum of a planet of mass m, moving around the Sun in a circular orbit is L, about the center of the Sun, its areal velocity is :

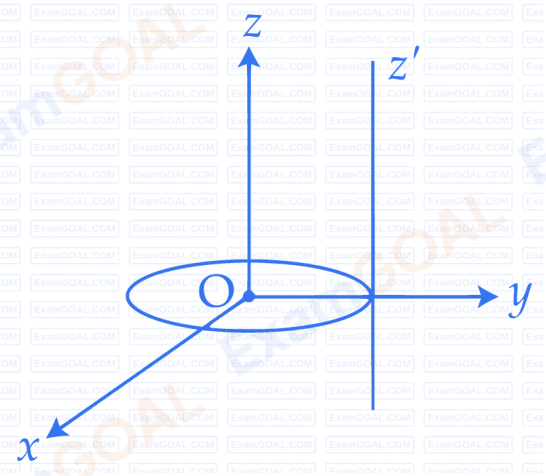
(A) $\frac{L}{m}$ (B) $\frac{4L}{m}$ (C) $\frac{L}{2m}$ (D) $\frac{2L}{m}$

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Q130. JEE Main 2018 (Online) 16th April Morning Slot

A thin circular disk is in the xy plane as shown in the figure. The ratio of its moment of inertia about z and z' axes will be :

Open solution

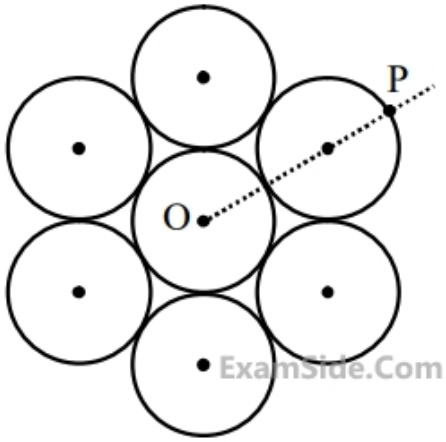


(A) 1 : 3 (B) 1 : 4 (C) 1 : 5 (D) 1 : 2

Q131. JEE Main 2018 (Offline)

Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is :

Open solution



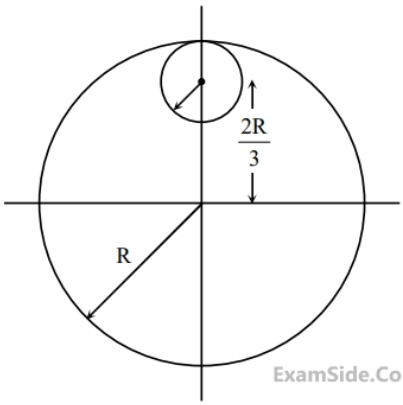
(A) $\frac{181}{2}MR^2$ (B) $\frac{55}{2}MR^2$ (C) $\frac{19}{2}MR^2$ (D) $\frac{73}{2}MR^2$

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Q132. JEE Main 2018 (Offline)

From a uniform circular disc of radius R and mass 9M, a small disc of radius R/3 is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is :

Open solution

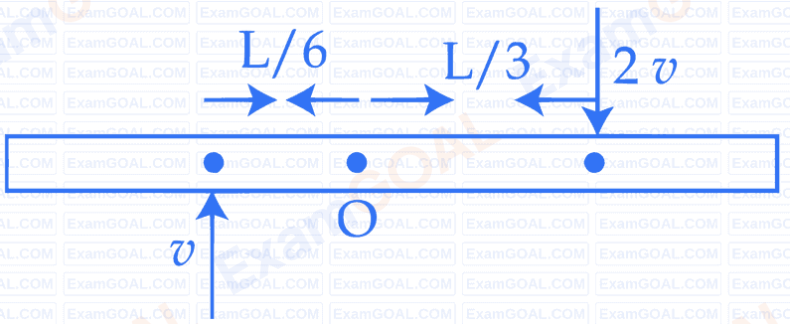


(A) $\frac{37}{9}MR^2$ (B) $4MR^2$ (C) $\frac{40}{9}MR^2$ (D) $10MR^2$

Q133. JEE Main 2018 (Online) 15th April Evening Slot

A thin uniform bar of length L and mass 8 m lies on a smooth horizontal table. Two point masses m and 2 m are moving in the same horizontal plane from opposite sides of the bar with speeds $2v$ and v respectively. The masses stick to the bar after collision at a distance $\frac{L}{3}$ and $\frac{L}{6}$ respectively from the center of the bar. If the bar starts rotating about its center of mass as a result of collision, the angular speed of the bar will be :

Open solution



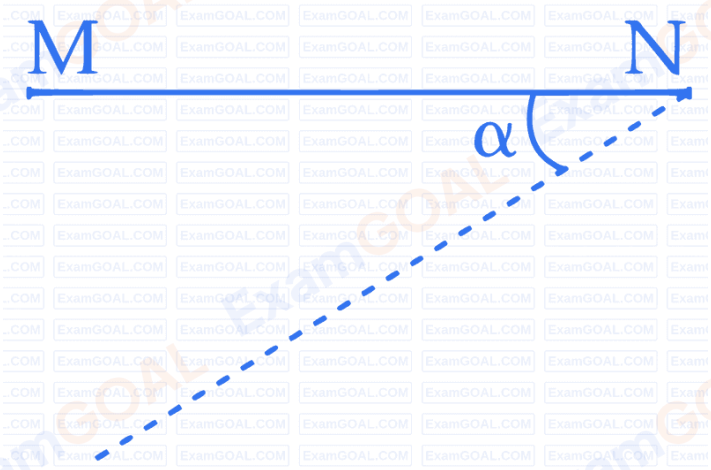
(A) $\frac{v}{5L}$ (B) $\frac{6v}{5L}$ (C) $\frac{3v}{5L}$ (D) $\frac{v}{6L}$

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Q134. JEE Main 2018 (Online) 15th April Evening Slot

A thin rod MN, free to rotate in the vertical plane about the fixed end N, is held horizontal. When the end M is released the speed of this end, when the rod makes an angle α with the horizontal, will be proportional to : (see figure)

Open solution

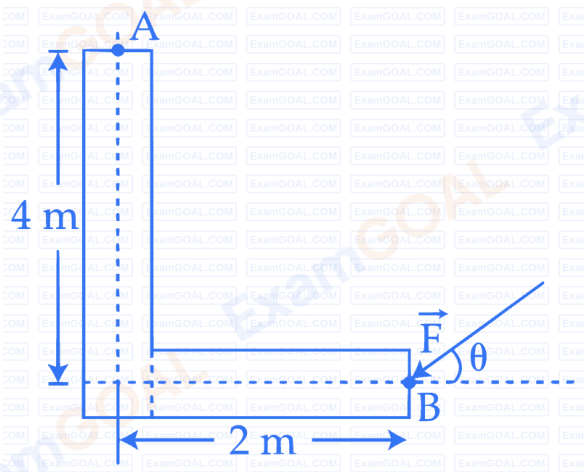


(A) $\sqrt{\sin\alpha}$ (B) $\sin\alpha$ (C) $\sqrt{\cos\alpha}$ (D) $\cos\alpha$

Q135. JEE Main 2018 (Online) 15th April Morning Slot

A force of $40\ N$ acts on a point B at the end of an L-shaped object, as shown in the figure. The angle θ that will produce maximum moment of the force about point A is given by :

Open solution

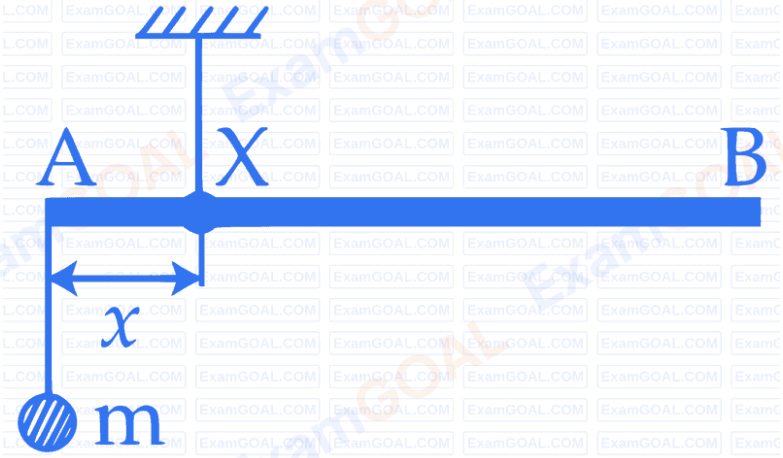


(A) $\tan\theta = \frac{1}{2}$ (B) $\tan\theta = 2$ (C) $\tan\theta = 4$ (D) $\tan\theta = \frac{1}{4}$

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Q136. JEE Main 2018 (Online) 15th April Morning Slot

Open solution



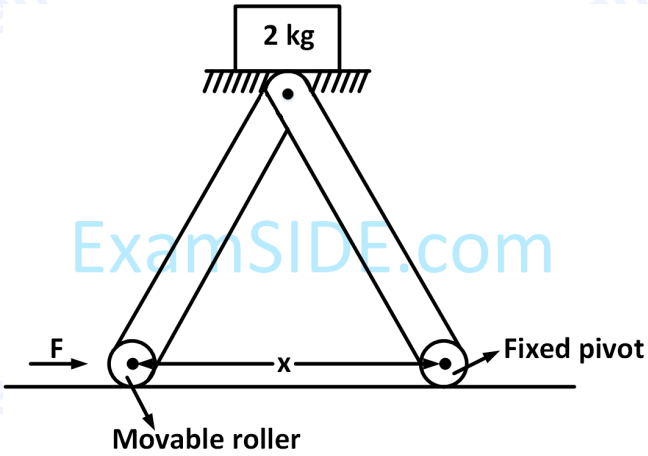
A uniform rod AB is suspended from a point X, at a variable distance x from A, as shown. To make the rod horizontal, a mass m is suspended from its end A. A set of (m, x) values is recorded. The appropriate variables that give a straight line, when plotted, are :

(A) m, x (B) $m, \frac{1}{x}$ (C) $m, \frac{1}{x^2}$ (D) m, x^2

Q137. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

The machine as shown has 2 rods of length1 m connected by a pivot at the top. The end of one rod is connected to the floor by a stationary pivot and the end of the other rod has a roller that rolls along the floor in a slot. As the roller goes back and forth, a 2 kg weight moves up and down. If the roller is moving towards right at a constant speed, the weight moves up with a :



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(A) Constant speed

(B) decreasing speed

(C) increasing speed

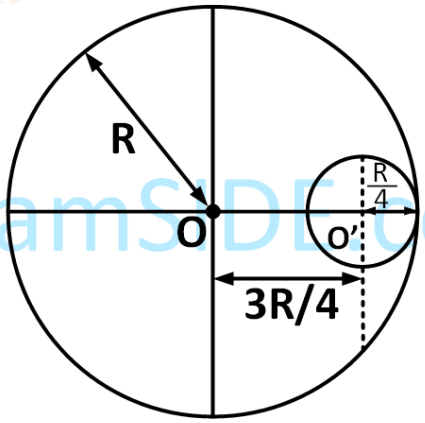
(D) speed which is $\frac{3}{4}$th of that of the roller when the weight is 0.4 m above the ground

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Q138. JEE Main 2017 (Online) 9th April Morning Slot

A circular hole of radius $\frac{R}{4}$ is made in a thin uniform disc having mass M and radius R, as shown in figure. The moment of inertia of the remaining portion of the disc about an axis passing through the point O and perpendicular to the plane of the disc is :

Open solution

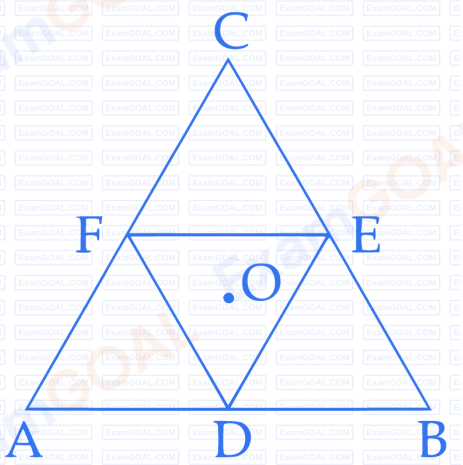


(A) $\frac{219\,MR^2}{256}$ (B) $\frac{237\,MR^2}{512}$ (C) $\frac{19\,MR^2}{512}$ (D) $\frac{197\,MR^2}{256}$

Q139. JEE Main 2017 (Online) 8th April Morning Slot

Moment of inertia of an equilateral triangular lamina ABC, about the axis passing through its centre O and perpendicular to its plane is I_o as shown in the figure. A cavity DEF is cut out from the lamina, where D, E, F are the mid points of the sides. Moment of inertia of the remaining part of lamina about the same axis is :

Open solution



(A) $\frac{7}{8}\,I_o$ (B) $\frac{15}{16}\,I_o$ (C) $\frac{3\,I_o}{4}$ (D) $\frac{31\,I_o}{32}$

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Q140. JEE Main 2017 (Online) 8th April Morning Slot

In a physical balance working on the principle of moments, when 5 mg weight is placed on the left pan, the beam becomes horizontal. Both the empty pans of the balance are of equal mass. Which of the following statements is **correct** ?

Open solution

(A) Left arm is longer than the right arm

(B) Both the arms are of same length

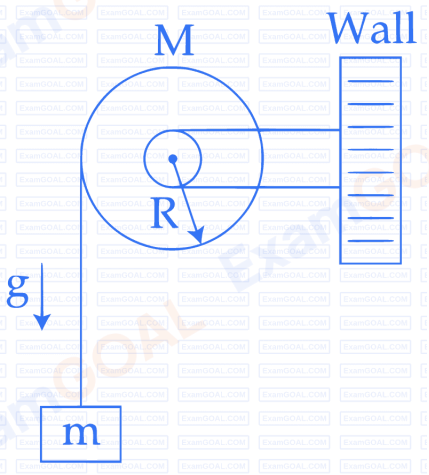
(C) Left arm is shorter than the right arm

(D) Every object that is weighed using this balance appears lighter than its actual weight.

Q141. JEE Main 2017 (Online) 8th April Morning Slot

A uniform disc of radius R and mass M is free to rotate only about its axis. A string is wrapped over its rim and a body of mass m is tied to the free end of the string as shown in the figure. The body is released from rest. Then the acceleration of the body is :

Open solution



(A) $\frac{2\ mg}{2\ m+M}$ (B) $\frac{2\ Mg}{2\ m+M}$ (C) $\frac{2\ mg}{2\ M+m}$ (D) $\frac{2\ Mg}{2\ M+m}$

Q142. JEE Main 2017 (Offline)

The moment of inertia of a uniform cylinder of length l and radius R about its perpendicular bisector is I. What is the ratio $\frac{l}{R}$ such that the moment of inertia is minimum?

Open solution

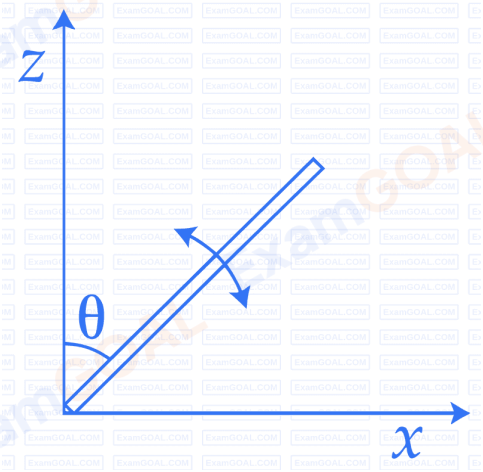
(A) $\frac{3}{\sqrt{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) 1

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Q143. JEE Main 2017 (Offline)

Open solution

A slender uniform rod of mass M and length l is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is

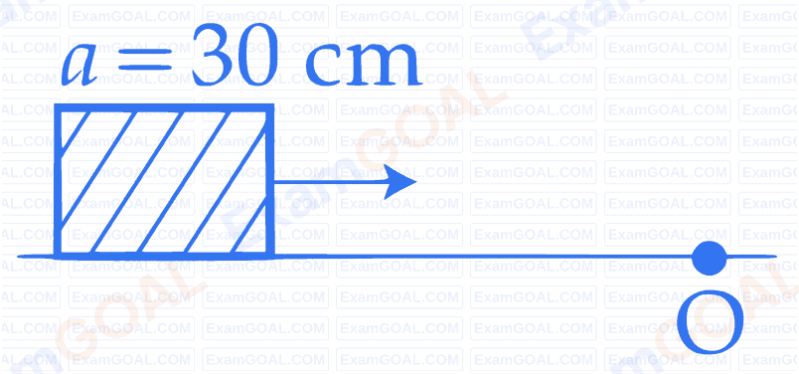


(A) $\frac{2g}{3l}\cos\theta$ (B) $\frac{3g}{2l}\sin\theta$ (C) $\frac{2g}{3l}\sin\theta$ (D) $\frac{3g}{3l}\sin\theta$

Q144. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

A cubical block of side 30 cm is moving with velocity 2 ms^{-1} on a smooth horizontal surface. The surface has a bump at a point O as shown in figure. The angular velocity (in rad/s) of the block immediately after it hits the bump, is :



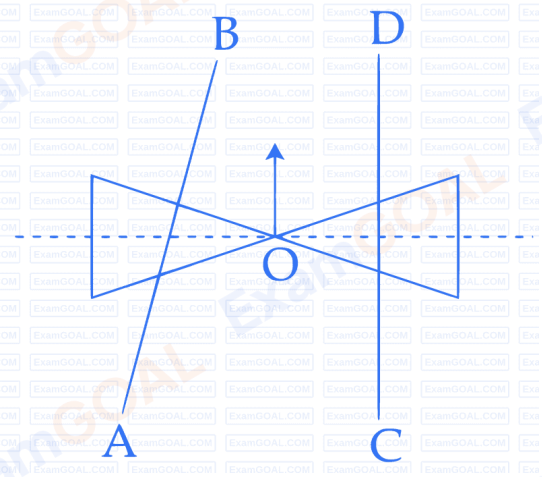
(A) 5.0 (B) 6.7 (C) 9.4 (D) 13.3

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Q145. JEE Main 2016 (Offline)

Open solution

A roller is made by joining together two cones at their vertices 0. It is kept on two rails AB and CD, which are placed asymmetrically (see figure), with its axis perpendicular to CD and its center O at the center of line joining AB and CD (see figure). It is given a light push so that it starts rolling with its center O moving parallel to CD in the direction shown. As it moves, the roller will tend to :



(A) go straight

(B) turn left and right alternately

(C) turn left

(D) turn right

Q146. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

A particle of mass 2 kg is on a smooth horizontal table and moves in a circular path of radius 0.6 m. The height of the table from the ground is 0.8 m. If the angular speed of the particle is 12 rad s^{-1}, the magnitude of its angular momentum about a point on the ground right under the center of the circle is :

(A) 8.64 kg m^2s^{-1} (B) 11.52 kg m^2s^{-1} (C) 14.4 kg m^2s^{-1} (D) 20.16 kg m^2s^{-1}

Q147. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

A uniform solid cylindrical roller of mass 'm' is being pulled on a horizontal surface with force F parallel to the surface and applied at its centre. If the acceleration of the cylinder is 'a' and it is rolling without slipping then the value of 'F' is :

(A) ma (B) 2 ma (C) $\frac{3}{2}$ ma (D) $\frac{5}{3}$ ma

Q148. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

Consider a thin uniform square sheet made of a rigid material. If its side is 'a', mass m and moment of inertia I about one of its diagonals, then :

(A) I > $\frac{ma^2}{12}$

(B) $\frac{ma^2}{24}$ < I < $\frac{ma^2}{12}$

(C) I = $\frac{ma^2}{12}$

(D) I = $\frac{ma^2}{24}$

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Q149. JEE Main 2015 (Offline)

From a solid sphere of mass M and radius R a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its center and perpendicular to one of its face is:

Open solution

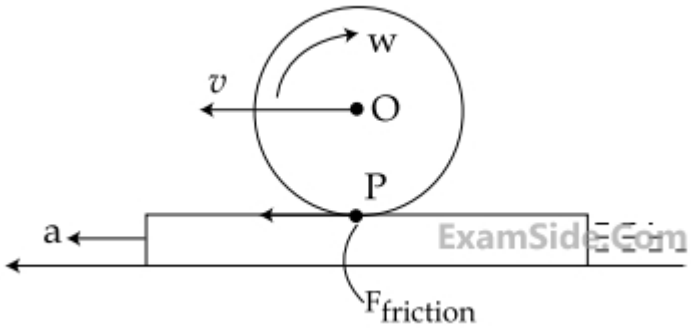
(A) $\frac{4MR^2}{9\sqrt{3}\,\pi}$ (B) $\frac{4MR^2}{3\sqrt{3}\,\pi}$

(C) $\frac{MR^2}{32\sqrt{2}\,\pi}$ (D) $\frac{MR^2}{16\sqrt{2}\,\pi}$

Q150. JEE Main 2014 (Online) 19th April Morning Slot

Consider a cylinder of mass M resting on a rough horizontal rug that is pulled out from under it with acceleration 'a' perpendicular to the axis of the cylinder. What is $F_{friction}$ at point P ? It is assumed that the cylinder does not slip.

Open solution



(A) Mg (B) Ma (C) $\frac{Ma}{2}$ (D) $\frac{Ma}{3}$

Q151. JEE Main 2014 (Online) 19th April Morning Slot

A ball of mass 160 g is thrown up at an angle of 60° to the horizontal at a speed of 10 ms^{-1}. The angular momentum of the ball at the highest point of the trajectory with respect to the point from which the ball is thrown is nearly (g = 10 ms^{-2})

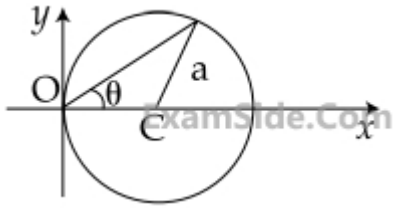
Open solution

(A) 1.73 kg m^2/s (B) 3.0 kg m^2/s (C) 3.46 kg m^2/s (D) 6.0 kg m^2/s

Q152. JEE Main 2014 (Online) 12th April Morning Slot

A particle is moving in a circular path of radius a, with a constant velocity v as shown in the figure. The center of circle is marked by 'C'. The angular momentum from the origin O can be written as :

Open solution



(A) $va(1 + \cos 2\theta)$ (B) $va(1 + \cos \theta)$

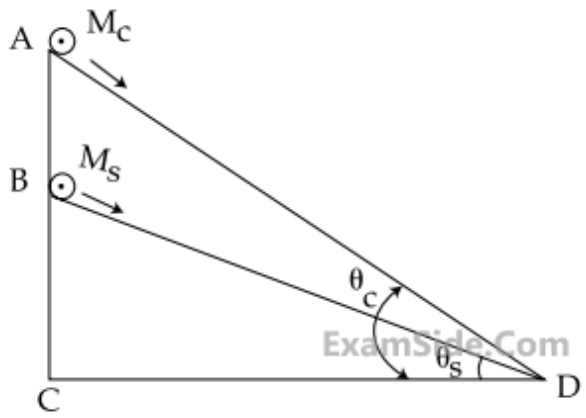
(C) $va \cos 2\theta$ (D) va

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Q153. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

A cylinder of mass M_C and sphere of mass M_S are placed at points A and B of two inclines, respectively. (See Figure). If they roll on the incline without slipping such that their accelerations are the same, then the ratio $\frac{\sin\theta_c}{\sin\theta_s}$ is :



(A) $\sqrt{\frac{8}{7}}$ (B) $\sqrt{\frac{15}{14}}$ (C) $\frac{8}{7}$ (D) $\frac{15}{14}$

Q154. JEE Main 2014 (Offline)

Open solution

A bob of mass m attached to an inextensible string of length l is suspended from a vertical support. The bob rotates in a horizontal circle with an angular speed $\omega\, rad/s$ about the vertical. About the point of suspension:

(A) angular momentum is conserved

(B) angular momentum changes in magnitude but not in direction.

(C) angular momentum changes in direction but not in magnitude.

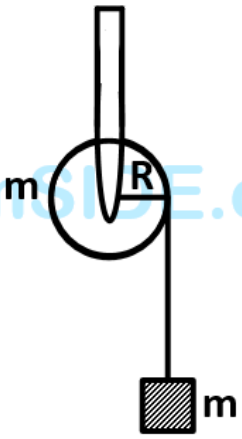
(D) angular momentum changes both in direction and magnitude.

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Q155. JEE Main 2014 (Offline)

A mass $'m'$ is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release?

Open solution



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(A) $\frac{2g}{3}$ (B) $\frac{g}{2}$ (C) $\frac{5g}{6}$ (D) g

Q156. JEE Main 2013 (Online) 25th April Morning Slot

A ring of mass M and radius R is rotating about its axis with angular velocity ω. Two identical bodies each of mass m are now gently attached at the two ends of a diameter of the ring. Because of this, the kinetic energy loss will be :

Open solution

(A) $\frac{m(M+2m)}{M}\omega^2R^2$ (B) $\frac{Mm}{(M+m)}\omega^2R^2$

(C) $\frac{Mm}{(M+2m)}\omega^2R^2$ (D) $\frac{(M+m)M}{(M+2m)}\omega^2R^2$

Q157. JEE Main 2013 (Online) 23th April Morning Slot

A particle of mass 2 kg is moving such that at time t, its position, in meter, is given by $\vec{r}(t) = 5\hat{i} - 2t^2\hat{j}$. The angular momentum of the particle at $t = 2s$ about the origin in kg m^2 s^{-1} is :

Open solution

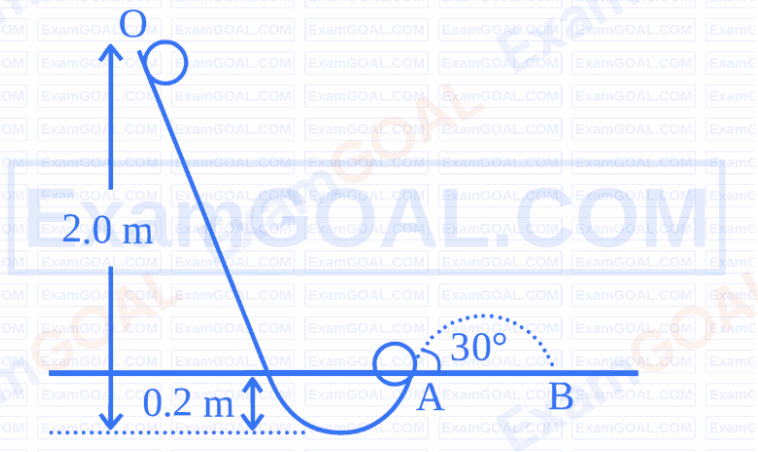
(A) $-80\hat{k}$ (B) $(10\hat{i} - 16\hat{j})$ (C) $-40\hat{k}$ (D) $40\hat{k}$

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Q158. JEE Main 2013 (Online) 22th April Morning Slot

A tennis ball (treated as hollow spherical shell) starting from O rolls down a hill. At point A, the ball becomes air borne leaving at an angle of 30° with the horizontal. The ball strikes the ground at B. What is the value of the distance AB? (Moment of inertia of a spherical shell of mass m and radius R about its diameter = $\frac{2}{3}mR^2$)

Open solution



(A) 1.87 m (B) 2.08 m (C) 1.57 m (D) 1.77 m

Q159. JEE Main 2013 (Online) 9th April Morning Slot

A bullet of mass 10 g and speed 500 m/s is fired into a door and gets embedded exactly at the centre of the door. The door is 1.0 m wide and weighs 12 kg. It is hinged at one end and rotates about a vertical axis practically without friction. The angular speed of the door just after the bullet embeds into it will be :

Open solution

(A) 6.25 rad/sec (B) 0.625 rad/sec (C) 3.35 rad/sec (D) 0.335 rad/sec

Q160. JEE Main 2013 (Offline)

A hoop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the center of the hoop is zero. What will be the velocity of the center of the hoop when it ceases to slip?

Open solution

(A) $\frac{r\omega_0}{4}$ (B) $\frac{r\omega_0}{3}$ (C) $\frac{r\omega_0}{2}$ (D) $r\omega_0$

Q161. AIEEE 2012 (Online) 26th May

A stone of mass m, tied to the end of a string, is whirled around in a circle on a horizontal frictionless table. The length of the string is reduced gradually keeping the angular momentum of the stone about the centre of the circle constant. Then, the tension in the string is given by $T = Ar^n$, where A is a constant, r is the instantaneous radius of the circle. The value of n is equal to

Open solution

(A) -1 (B) -2 (C) -4 (D) -3

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Q162. AIEEE 2012 (Online) 26th May

A thick-walled hollow sphere has outside radius R_0. It rolls down an incline without slipping and its speed at the bottom is v_0. Now the incline is waxed, so that it is practically frictionless and the sphere is observed to slide down (without any rolling). Its speed at the bottom is observed to be $5v_0/4$. The radius of gyration of the hollow sphere about an axis through its centre is

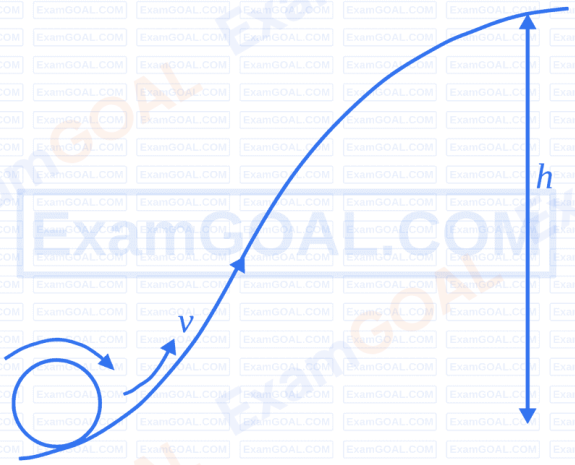
Open solution

(A) $3R_0/2$ (B) $3R_0/4$ (C) $9R_0/16$ (D) $3R_0$

Q163. AIEEE 2012 (Online) 12th May

A solid sphere is rolling on a surface as shown in figure, with a translational velocity $v\,\mathrm{m\,s^{-1}}$. If it is to climb the inclined surface continuing to roll without slipping, then minimum velocity for this to happen is

Open solution



(A) $\sqrt{2gh}$ (B) $\sqrt{\frac{7}{5}gh}$ (C) $\sqrt{\frac{7}{2}gh}$ (D) $\sqrt{\frac{10}{7}gh}$

Q164. AIEEE 2012 (Online) 12th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: When moment of inertia I of a body rotating about an axis with angular speed ω increases, its angular momentum L is unchanged but the kinetic energy K increases if there is no torque applied on it.
Statement 2: $L = I\omega$, kinetic energy of rotation $= \frac{1}{2}I\omega^2$

Open solution

(A) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(B) Statement 1 is false, Statement 2 is true.

(C) Statement 1 is true, Statement 2 is true, Statement 2 is correct explanation of the Statement 1.

(D) Statement 1 is true, Statement 2 is false.

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Q165. AIEEE 2012 (Online) 7th May

A circular hole of diameter R is cut from a disc of mass M and radius R; the circumference of the cut passes through the centre of the disc. The moment of inertia of the remaining portion of the disc about an axis perpendicular to the disc and passing through its centre is

(A) $\left(\frac{15}{32}\right)MR^2$ (B) $\left(\frac{1}{8}\right)MR^2$ (C) $\left(\frac{3}{8}\right)MR^2$ (D) $\left(\frac{13}{32}\right)MR^2$

Open solution

Q166. AIEEE 2012 (Online) 7th May

A solid sphere having mass m and radius r rolls down an inclined plane. Then its kinetic energy is

(A) $\frac{5}{7}$ rotational and $\frac{2}{7}$ translational

(B) $\frac{2}{7}$ rotational and $\frac{5}{7}$ translational

(C) $\frac{2}{5}$ rotational and $\frac{3}{5}$ translational

(D) $\frac{1}{2}$ rotational and $\frac{1}{2}$ translational

Open solution

Q167. AIEEE 2011

A mass m hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass m and radius R. Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass m, if the string does not slip on the pulley, is:

(A) g (B) $\frac{2}{3}g$ (C) $\frac{g}{3}$ (D) $\frac{3}{2}g$

Open solution

Q168. AIEEE 2011

A thin horizontal circular disc is rotating about a vertical axis passing through its center. An insect is at rest at a point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc.

(A) continuously decreases (B) continuously increases

(C) first increases and then decreases (D) remains unchanged

Open solution

Q169. AIEEE 2011

A pulley of radius $2\ m$ is rotated about its axis by a force $F = (20t - 5t^2)$ newton (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is $10kg\text{-}m^2$ the number of rotation made by the pulley before its direction of motion is reversed, is:

(A) more than 3 but less than 6 (B) more than 6 but less than 9

(C) more than 9 (D) less than 3

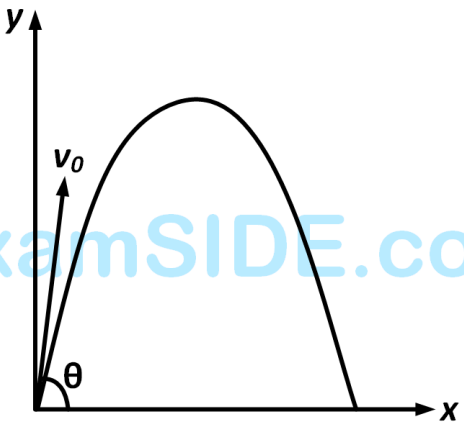
Open solution

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Q170. AIEEE 2010

A small particle of mass m is projected at an angle θ with the x-axis with an initial velocity v_0 in the x-y plane as shown in the figure. At a time $t < \frac{v_0 \sin\theta}{g}$, the angular momentum of the particle is .....,

Open solution



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where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along x, y and z-axis respectively.

(A) $-mg\,v_0 t^2 \cos\theta\hat{j}$

(B) $mg\,v_0 t \cos\theta\hat{k}$

(C) $-\frac{1}{2}mg\,v_0 t^2 \cos\,\theta\hat{k}$

(D) $\frac{1}{2}mg\,v_0 t^2 \cos\theta\hat{i}$

Q171. AIEEE 2009

A thin uniform rod of length l and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω. Its center of mass rises to a maximum height of:

Open solution

(A) $\frac{1}{6}\,\frac{l\omega}{g}$

(B) $\frac{1}{2}\,\frac{l^2\omega^2}{g}$

(C) $\frac{1}{6}\,\frac{l^2\omega^2}{g}$

(D) $\frac{1}{3}\,\frac{l^2\omega^2}{g}$

Q172. AIEEE 2008

Consider a uniform square plate of side $'a'$ and mass $'m'$. The moment of inertia of this plate about an axis perpendicular to its plane and passing through one of its corners is

Open solution

(A) $\frac{5}{6}ma^2$

(B) $\frac{1}{12}ma^2$

(C) $\frac{7}{12}ma^2$

(D) $\frac{2}{3}ma^2$

Q173. AIEEE 2007

A round uniform body of radius R, mass M and moment of inertia I rolls down (without slipping) an inclined plane making an angle θ with the horizontal. Then its acceleration is

Open solution

(A) $\frac{g\,\sin\theta}{1-MR^2/I}$

(B) $\frac{g\,\sin\theta}{1+I/MR^2}$

(C) $\frac{g\,\sin\theta}{1+MR^2/I}$

(D) $\frac{g\,\sin\theta}{1-I/MR^2}$

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Q174. AIEEE 2007

Angular momentum of the particle rotating with a central force is constant due to

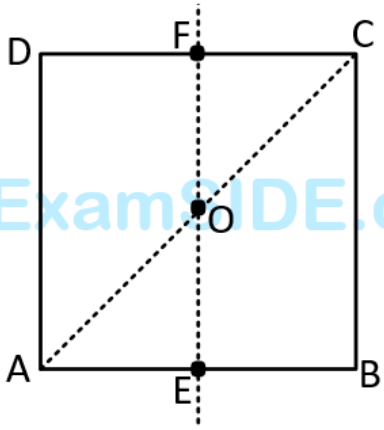
(A) constant torque (B) constant force

(C) constant linear momentum (D) zero torque

Open solution

Q175. AIEEE 2007

For the given uniform square lamina $ABCD$, whose center is O,



(A) $I_{AC} = \sqrt{2}\ I_{EF}$ (B) $\sqrt{2}I_{AC} = I_{EF}$ (C) $I_{AD} = 3I_{EF}$ (D) $I_{AC} = I_{EF}$

Open solution

Q176. AIEEE 2006

Four point masses, each of value m, are placed at the corners of a square $ABCD$ of side l. The moment of inertia of this system about an axis passing through A and parallel to BD is

(A) $2ml^2$ (B) $\sqrt{3}ml^2$ (C) $3ml^2$ (D) ml^2

Open solution

Q177. AIEEE 2006

A thin circular ring of mass m and radius R is rotating about its axis with a constant angular velocity ω. Two objects each of mass M are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with an angular velocity $\omega' =$

(A) $\frac{\omega(m+2M)}{m}$ (B) $\frac{\omega(m-2M)}{(m+2M)}$

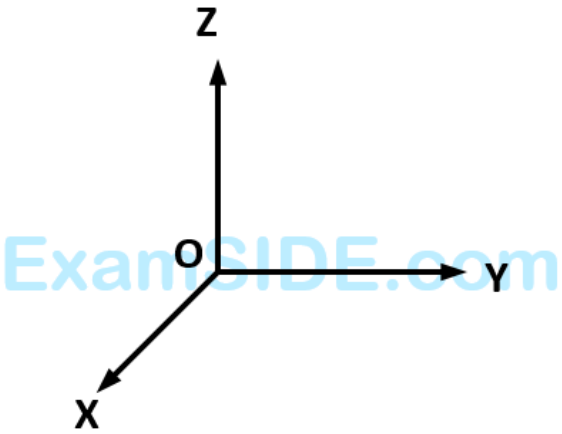
(C) $\frac{\omega m}{(m+M)}$ (D) $\frac{\omega m}{(m+2M)}$

Open solution

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Q178. AIEEE 2006

A force of $-F\hat{k}$ acts on O, the origin of the coordinate system. The torque about the point $(1, -1)$ is



Open solution

(A) $F\left(\hat{i}-\hat{j}\right)$ (B) $-F\left(\hat{i}+\hat{j}\right)$

(C) $F\left(\hat{i}+\hat{j}\right)$ (D) $-F\left(\hat{i}-\hat{j}\right)$

Q179. AIEEE 2005

An annular ring with inner and outer radii R_1 and R_2 is rolling without slipping with a uniform angular speed. The ratio of the forces experienced by the two particles situated on the inner and outer parts of the ring, $\frac{F_1}{F_2}$ is

Open solution

(A) $\left(\frac{R_1}{R_2}\right)^2$ (B) $\frac{R_2}{R_1}$

(C) $\frac{R_1}{R_2}$ (D) 1

Q180. AIEEE 2005

The moment of inertia of a uniform semicircular disc of mass M and radius r about a line perpendicular to the plane of the disc through the center is

Open solution

(A) $\frac{2}{5}Mr^2$ (B) $\frac{1}{4}Mr$ (C) $\frac{1}{2}Mr^2$ (D) Mr^2

Q181. AIEEE 2004

A solid sphere is rotating in free space. If the radius of the sphere is increased keeping mass same which one of the following will not be affected ?

Open solution

(A) Angular velocity (B) Angular momentum

(C) Moment of inertia (D) Rotational kinetic energy

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Q182. AIEEE 2004

One solid sphere A and another hollow sphere B are of same mass and same outer radii. Their moment of inertia about their diameters are respectively I_A and I_B such that

(A) $I_A < I_B$

(B) $I_A > I_B$

(C) $I_A = I_B$

(D) $\frac{I_A}{I_B} = \frac{d_A}{d_B}$ where d_A and d_B are their densities.

Open solution

Q183. AIEEE 2003

A circular disc X of radius R is made from an iron plate of thickness t, and another disc Y of radius $4\,R$ is made from an iron plate of thickness $\frac{t}{4}$. Then the relation between the moment of inertia I_X and I_Y is

(A) $I_Y = 32I_X$ (B) $I_Y = 16I_X$ (C) $I_Y = I_X$ (D) $I_Y = 64I_X$

Open solution

Q184. AIEEE 2003

A particle performing uniform circular motion has angular momentum L. If its angular frequency is doubled & its kinetic energy halved, then the new angular momentum is

(A) $\frac{L}{4}$ (B) $2L$ (C) $4L$ (D) $\frac{L}{2}$

Open solution

Q185. AIEEE 2003

Let $\vec{F}$ be the force acting on a particle having position vector $\vec{r}$, and $\vec{\tau}$ be the torque of this force about the origin. Then

(A) $\vec{r}.\vec{\tau} = 0$ and $\vec{F}.\vec{\tau} \neq 0$

(B) $\vec{r}.\vec{\tau} \neq 0$ and $\vec{F}.\vec{\tau} = 0$

(C) $\vec{r}.\vec{\tau} \neq 0$ and $\vec{F}.\vec{\tau} \neq 0$

(D) $\vec{r}.\vec{\tau} = 0$ and $\vec{F}.\vec{\tau} = 0$

Open solution

Q186. AIEEE 2002

Initial angular velocity of a circular disc of mass M is ω_1. Then two small spheres of mass m are attached gently to diametrically opposite points on the edge of the disc. What is the final angular velocity of the disc?

(A) $\left(\frac{M+m}{M}\right)\omega_1$ (B) $\left(\frac{M+m}{m}\right)\omega_1$

(C) $\left(\frac{M}{M+4m}\right)\omega_1$ (D) $\left(\frac{M}{M+2m}\right)\omega_1$

Open solution

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Q187. AIEEE 2002

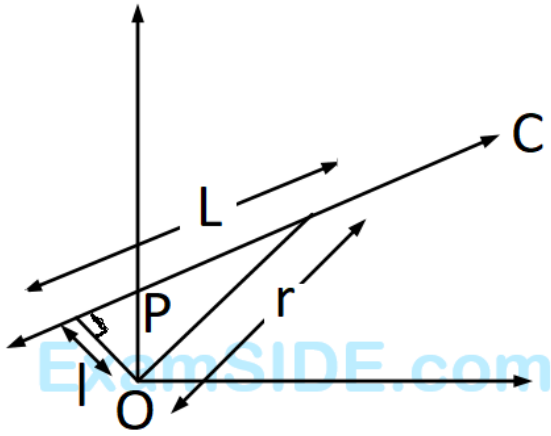
A solid sphere, a hollow sphere and a ring are released from top of an inclined plane (frictionless) so that they slide down the plane. Then maximum acceleration down the plane is for (no rolling)

(A) solid sphere (B) hollow sphere (C) ring (D) all same

Open solution

Q188. AIEEE 2002

A particle of mass m moves along line PC with velocity v as shown. What is the angular momentum of the particle about P?



(A) mvL (B) mvl (C) mvr (D) zero

Open solution

Q189. AIEEE 2002

Moment of inertia of a circular wire of mass M and radius R about its diameter is

(A) $\frac{MR^2}{2}$ (B) MR^2 (C) $2MR^2$ (D) $\frac{MR^2}{4}$

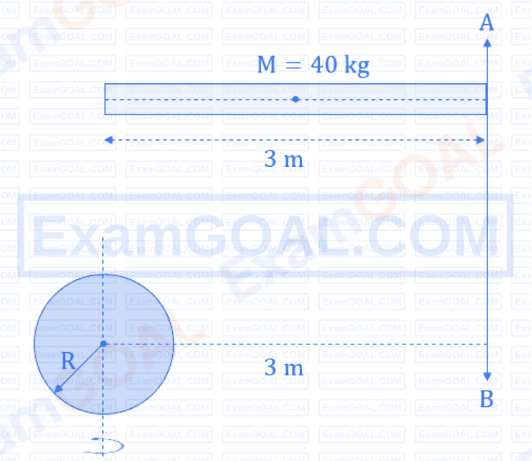
Open solution

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Numerical 96 questions

Q190. JEE Main 2026 (Online) 2nd April Evening Shift

Moment of inertia about an axis *AB* for a rod of mass 40 kg and length 3 m is same as that of a solid sphere of mass of 10 kg and radius R about an axis parallel to *AB* axis with separation of 3 m as shown in figure below. The value of R is given as $\sqrt{\frac{\alpha}{2}}$. The value of α is _____.



Open solution

Q191. JEE Main 2026 (Online) 28th January Evening Shift

A fly wheel having mass 3 kg and radius 5 m is free to rotate about a horizontal axis. A string having negligible mass is wound around the wheel and the loose end of the string is connected to 3 kg mass. The mass is kept at rest initially and released. Kinetic energy of the wheel when the mass descends by 3 m is _____ J. ($g = 10\ \mathrm{m/s^2}$)

Open solution

Q192. JEE Main 2026 (Online) 28th January Morning Shift

A solid sphere of radius 10 cm is rotating about an axis which is at a distance 15 cm from its centre. The radius of gyration about this axis is $\sqrt{n}$ cm. The value of n is

Open solution

Q193. JEE Main 2026 (Online) 24th January Evening Shift

A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1. A small co-centric cylinder of length $L/2$ and radius $R/3$ carved from this cylinder has moment of inertia about its axis equals to I_2. The ratio I_1/I_2 is ____ .

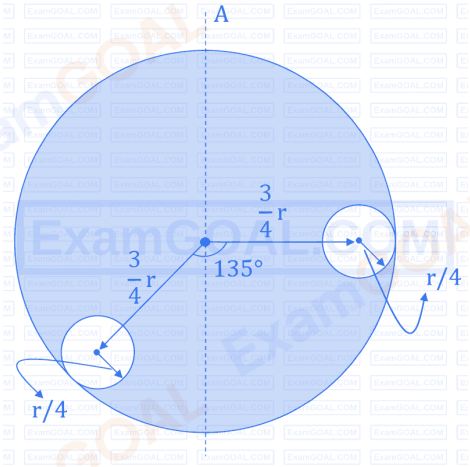
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Q194. JEE Main 2026 (Online) 23rd January Evening Shift

Suppose there is a uniform circular disc of mass M kg and radius r m shown in figure. The shaded regions are cut out from the disc. The moment of inertia of the remainder about the axis A of the disc is given by $\frac{x}{256}Mr^2$. The value of x is ____ .

Open solution



Q195. JEE Main 2026 (Online) 22nd January Evening Shift

Two masses m and 2 m are connected by a light string going over a pulley (disc) of mass 30 m with radius $r = 0.1$ m. The pulley is mounted in a vertical plane and it is free to rotate about its axis. The 2 m mass is released from rest and its speed when it has descended through a height of 3.6 m is ____ m/s. (Assume string does not slip and g $= 10$ m/s^2)

Open solution

Q196. JEE Main 2026 (Online) 22nd January Morning Shift

A circular disc has radius R_1 and thickness T_1. Another circular disc made of the same material has radius R_2 and thickness T_2. If the moment of inertia of both discs are same and $\frac{R_1}{R_2} = 2$ then $\frac{T_1}{T_2} = \frac{1}{\alpha}$. The value of α is ____ .

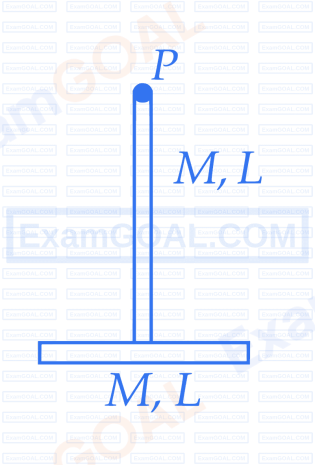
Open solution

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Q197. JEE Main 2026 (Online) 21st January Morning Shift

Two identical thin rods of mass M kg and length L m are connected as shown in figure. Moment of inertia of the combined rod system about an axis passing through point P and perpendicular to the plane of the rods is $\frac{x}{12}ML^2$ kg m^2. The value of x is ____ .

Open solution



Q198. JEE Main 2025 (Online) 8th April Evening Shift

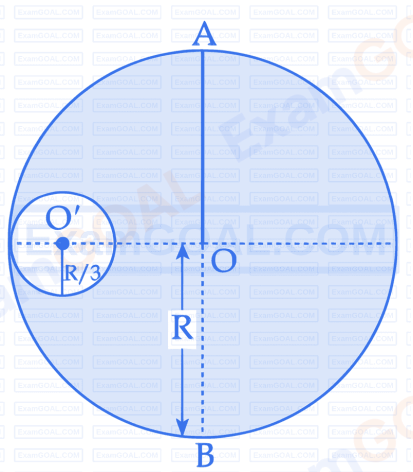
A thin solid disk of 1 kg is rotating along its diameter axis at the speed of 1800 rpm . By applying an external torque of 25π Nm for 40 s , the speed increases to 2100 rpm . The diameter of the disk is _____ m.

Open solution

Q199. JEE Main 2025 (Online) 7th April Evening Shift

Let M and R be the mass and radius of a disc. A small disc of radius $R/3$ is removed from the bigger disc as shown in figure. The moment of inertia of remaining part of bigger disc about an axis AB passing through the centre O and perpendicular to the plane of disc is $\frac{4}{x}MR^2$. The value of x is _____.

Open solution

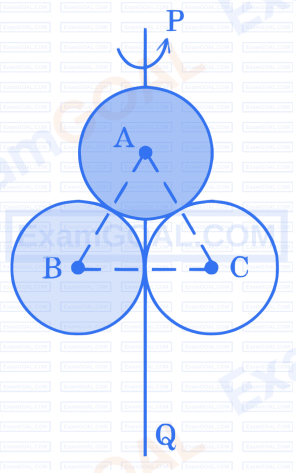


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Q200. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

A, B and C are disc, solid sphere and spherical shell respectively with same radii and masses. These masses are placed as shown in figure.



The moment of inertia of the given system about PQ axis is $\frac{x}{15}I$, where I is the moment of inertia of the disc about its diameter. The value of x is _____.

Q201. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

A solid sphere with uniform density and radius R is rotating initially with constant angular velocity (ω_1) about its diameter. After some time during the rotation it starts losing mass at a uniform rate, with no change in its shape. The angular velocity of the sphere when its radius become $R/2$ is $x\omega_1$. The value of x is _____.

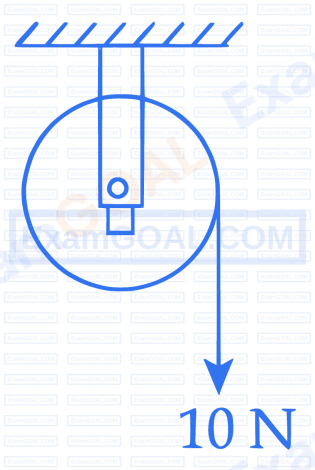
Q202. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

A circular ring and a solid sphere having same radius roll down on an inclined plane from rest without slipping. The ratio of their velocities when reached at the bottom of the plane is $\sqrt{\frac{x}{5}}$ where $x =$ _____.

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Q203. JEE Main 2025 (Online) 2nd April Evening Shift



Open solution

A wheel of radius 0.2 m rotates freely about its center when a string that is wrapped over its rim is pulled by force of 10 N as shown in figure. The established torque produces an angular acceleration of $2 \mathrm{rad/s^2}$. Moment of intertia of the wheel is_____ $\mathrm{kg\ m^2}$. (Acceleration due to gravity $= 10\ \mathrm{m/s^2}$)

Q204. JEE Main 2025 (Online) 29th January Morning Shift

The coordinates of a particle with respect to origin in a given reference frame is (1, 1, 1) meters. If a force of $\vec{F} = \hat{i} - \hat{j} + \hat{k}$ acts on the particle, then the magnitude of torque (with respect to origin) in z-direction is _____.

Open solution

Q205. JEE Main 2025 (Online) 28th January Morning Shift

Two iron solid discs of negligible thickness have radii R_1 and R_2 and moment of inertia I_1 and I_2, respectively. For $R_2 = 2R_1$, the ratio of I_1 and I_2 would be $1/x$, where $\mathrm{x} =$ _____ .

Open solution

Q206. JEE Main 2025 (Online) 28th January Morning Shift

The moment of inertia of a solid disc rotating along its diameter is 2.5 times higher than the moment of inertia of a ring rotating in similar way. The moment of inertia of a solid sphere which has same radius as the disc and rotating in similar way, is n times higher than the moment of inertia of the given ring. Here, $\mathrm{n} =$ _____ Consider all the bodies have equal masses.

Open solution

Q207. JEE Main 2025 (Online) 22nd January Morning Shift

The position vectors of two 1 kg particles, (A) and (B), are given by $\overrightarrow{\mathrm{r_A}} = \left(\alpha_1 \mathrm{t}^2\hat{i} + \alpha_2 \mathrm{t}\hat{j} + \alpha_3 \mathrm{t}\hat{k}\right)\mathrm{m}$ and $\overrightarrow{\mathrm{r_B}} = \left(\beta_1 \mathrm{t}\hat{i} + \beta_2 \mathrm{t}^2\hat{j} + \beta_3 \mathrm{t}\hat{k}\right)\mathrm{m}$, respectively; $\left(\alpha_1 = 1\ \mathrm{m/s^2}, \alpha_2 = 3n\ \mathrm{m/s}, \alpha_3 = 2\ \mathrm{m/s}, \beta_1 = 2\ \mathrm{m/s}, \beta_2 = -1\ \mathrm{m/s^2}, \beta_3 = 4p\ \mathrm{m/s}\right)$, where t is time, n and p are constants. At $t = 1\ \mathrm{s}$, $\left|\overrightarrow{V_A}\right| = \left|\overrightarrow{V_B}\right|$ and velocities $\vec{V}_A$ and $\vec{V}_B$ of the particles are orthogonal to each other. At $t = 1\ \mathrm{s}$, the magnitude of angular momentum of particle (A) with respect to the position of particle (B) is $\sqrt{\mathrm{L}}\mathrm{kgm^2\ s^{-1}}$. The value of L is _____.

Open solution

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Q208. JEE Main 2024 (Online) 9th April Evening Shift

A circular disc reaches from top to bottom of an inclined plane of length l. When it slips down the plane, it takes t s. When it rolls down the plane then it takes $\left(\frac{\alpha}{2}\right)^{1/2} t$ s, where α is _____.

Open solution

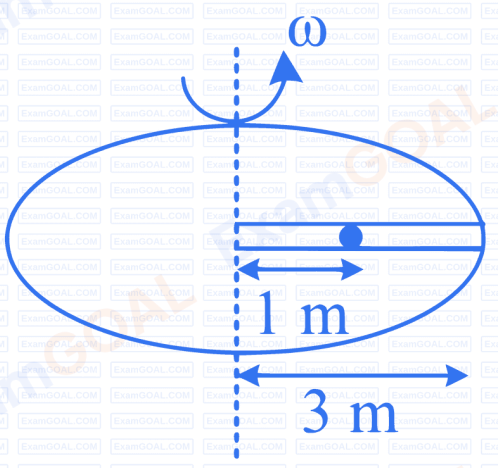
Q209. JEE Main 2024 (Online) 9th April Morning Shift

A string is wrapped around the rim of a wheel of moment of inertia $0.40\ \text{kgm}^2$ and radius $10\ \text{cm}$. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a force of $40\ \text{N}$. The angular velocity of the wheel after $10\ \text{s}$ is $x\ \text{rad/s}$, where x is _____.

Open solution

Q210. JEE Main 2024 (Online) 8th April Evening Shift

A circular table is rotating with an angular velocity of $\omega\ \text{rad/s}$ about its axis (see figure). There is a smooth groove along a radial direction on the table. A steel ball is gently placed at a distance of $1\ \text{m}$ on the groove. All the surfaces are smooth. If the radius of the table is $3\ \text{m}$, the radial velocity of the ball w.r.t. the table at the time ball leaves the table is $x\sqrt{2}\omega\ \text{m/s}$, where the value of x is _____.



Open solution

Q211. JEE Main 2024 (Online) 6th April Evening Shift

Three balls of masses $2\ \text{kg}, 4\ \text{kg}$ and $6\ \text{kg}$ respectively are arranged at centre of the edges of an equilateral triangle of side $2\ \text{m}$. The moment of inertia of the system about an axis through the centroid and perpendicular to the plane of triangle, will be _____ kg m^2.

Open solution

Q212. JEE Main 2024 (Online) 6th April Morning Shift

If the radius of earth is reduced to three-fourth of its present value without change in its mass then value of duration of the day of earth will be _____ hours 30 minutes.

Open solution

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Q213. JEE Main 2024 (Online) 5th April Evening Shift

A hollow sphere is rolling on a plane surface about its axis of symmetry. The ratio of rotational kinetic energy to its total kinetic energy is $\frac{x}{5}$. The value of x is _____.

Open solution

Q214. JEE Main 2024 (Online) 4th April Morning Shift

A solid sphere and a hollow cylinder roll up without slipping on same inclined plane with same initial speed v. The sphere and the cylinder reach upto maximum heights h_1 and h_2 respectively, above the initial level. The ratio $h_1 : h_2$ is $\frac{n}{10}$. The value of n is _____.

Open solution

Q215. JEE Main 2024 (Online) 1st February Evening Shift

A uniform rod AB of mass $2\ \text{kg}$ and length $30\ \text{cm}$ is at rest on a smooth horizontal surface. An impulse of force $0.2\ \text{Ns}$ is applied to end B. The time taken by the rod to turn through at right angles will be $\frac{\pi}{x}$ s, where $x =$ _____ .

Open solution

Q216. JEE Main 2024 (Online) 31st January Evening Shift

A body of mass 'm' is projected with a speed 'u' making an angle of 45° with the ground. The angular momentum of the body about the point of projection, at the highest point is expressed as $\frac{\sqrt{2}mu^3}{Xg}$. The value of 'X' is _____.

Open solution

Q217. JEE Main 2024 (Online) 31st January Evening Shift

Two identical spheres each of mass $2\ \text{kg}$ and radius $50\ \text{cm}$ are fixed at the ends of a light rod so that the separation between the centers is $150\ \text{cm}$. Then, moment of inertia of the system about an axis perpendicular to the rod and passing through its middle point is $\frac{x}{20}\ \text{kgm}^2$, where the value of x is _____.

Open solution

Q218. JEE Main 2024 (Online) 31st January Morning Shift

A solid circular disc of mass $50\ \text{kg}$ rolls along a horizontal floor so that its center of mass has a speed of $0.4\ \text{m/s}$. The absolute value of work done on the disc to stop it is _____ J.

Open solution

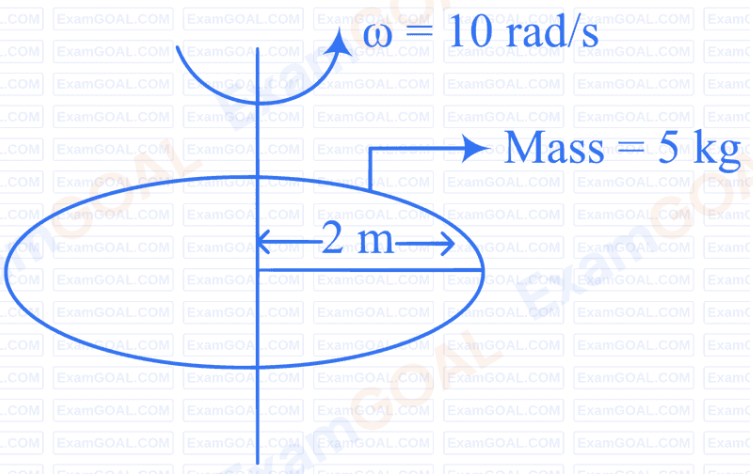
Q219. JEE Main 2024 (Online) 30th January Evening Shift

Two discs of moment of inertia $I_1 = 4\ \text{kg m}^2$ and $I_2 = 2\ \text{kg m}^2$, about their central axes & normal to their planes, rotating with angular speeds $10\ \text{rad/s}$ & $4\ \text{rad/s}$ respectively are brought into contact face to face with their axes of rotation coincident. The loss in kinetic energy of the system in the process is _____ J.

Open solution

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Q220. JEE Main 2024 (Online) 30th January Morning Shift



Open solution

Consider a Disc of mass $5\ \mathrm{kg}$, radius $2\ \mathrm{m}$, rotating with angular velocity of $10\ \mathrm{rad/s}$ about an axis perpendicular to the plane of rotation. An identical disc is kept gently over the rotating disc along the same axis. The energy dissipated so that both the discs continue to rotate together without slipping is _____ J.

Q221. JEE Main 2024 (Online) 29th January Evening Shift

A body of mass $5\ \mathrm{kg}$ moving with a uniform speed $3\sqrt{2}\ \mathrm{ms}^{-1}$ in $X-Y$ plane along the line $y = x + 4$. The angular momentum of the particle about the origin will be _____ $\mathrm{kg\ m^2\ s^{-1}}$.

Open solution

Q222. JEE Main 2024 (Online) 29th January Morning Shift

A cylinder is rolling down on an inclined plane of inclination 60°. Its acceleration during rolling down will be $\frac{x}{\sqrt{3}} m/s^2$, where $x =$ _____ (use $\mathrm{g} = 10\ \mathrm{m/s^2}$).

Open solution

Q223. JEE Main 2024 (Online) 27th January Evening Shift

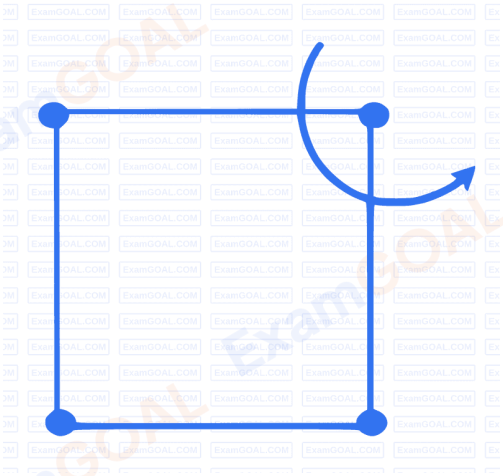
A ring and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of both bodies are identical and the ratio of their kinetic energies is $\frac{7}{x}$, where x is _____.

Open solution

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Q224. JEE Main 2024 (Online) 27th January Morning Shift

Four particles each of mass $1\ \mathrm{kg}$ are placed at four corners of a square of side $2\ \mathrm{m}$. Moment of inertia of system about an axis perpendicular to its plane and passing through one of its vertex is _____ kgm^2.



Open solution

Q225. JEE Main 2023 (Online) 15th April Morning Shift

A solid sphere and a solid cylinder of same mass and radius are rolling on a horizontal surface without slipping. The ratio of their radii of gyration respectively $\left(k_{\text{sph}} : k_{\text{cyl}}\right)$ is $2 : \sqrt{x}$. The value of x is _____ .

Open solution

Q226. JEE Main 2023 (Online) 13th April Evening Shift

A light rope is wound around a hollow cylinder of mass 5 kg and radius 70 cm. The rope is pulled with a force of 52.5 N. The angular acceleration of the cylinder will be _____ rad s^{-2}.

Open solution

Q227. JEE Main 2023 (Online) 13th April Morning Shift

A solid sphere is rolling on a horizontal plane without slipping. If the ratio of angular momentum about axis of rotation of the sphere to the total energy of moving sphere is $\pi : 22$ then, the value of its angular speed will be _____ $\mathrm{rad/s}$.

Open solution

Q228. JEE Main 2023 (Online) 12th April Morning Shift

For a rolling spherical shell, the ratio of rotational kinetic energy and total kinetic energy is $\frac{x}{5}$. The value of x is _____.

Open solution

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Q229. JEE Main 2023 (Online) 11th April Evening Shift

A circular plate is rotating in horizontal plane, about an axis passing through its center and perpendicular to the plate, with an angular velocity ω. A person sits at the center having two dumbbells in his hands. When he stretches out his hands, the moment of inertia of the system becomes triple. If E be the initial Kinetic energy of the system, then final Kinetic energy will be $\frac{E}{x}$. The value of x is

Open solution

Q230. JEE Main 2023 (Online) 11th April Morning Shift

A solid sphere of mass $500\ \mathrm{g}$ and radius $5\ \mathrm{cm}$ is rotated about one of its diameters with angular speed of $10\ \mathrm{rad\ s^{-1}}$. If the moment of inertia of the sphere about its tangent is $x \times 10^{-2}$ times its angular momentum about the diameter. Then the value of x will be _____.

Open solution

Q231. JEE Main 2023 (Online) 10th April Evening Shift

A force of $-\mathrm{P}\hat{\mathrm{k}}$ acts on the origin of the coordinate system. The torque about the point $(2, -3)$ is $\mathrm{P}(a\hat{i} + b\hat{j})$. The ratio of $\frac{a}{b}$ is $\frac{x}{2}$. The value of x is _____.

Open solution

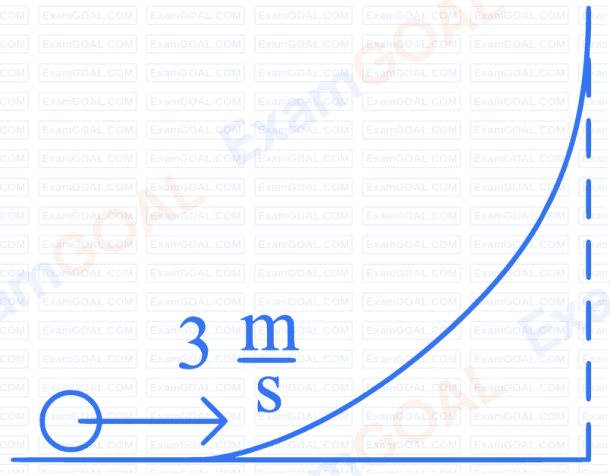
Q232. JEE Main 2023 (Online) 10th April Morning Shift

If the earth suddenly shrinks to $\frac{1}{64}$th of its original volume with its mass remaining the same, the period of rotation of earth becomes $\frac{24}{x}$h. The value of x is _____.

Open solution

Q233. JEE Main 2023 (Online) 8th April Evening Shift

A hollow spherical ball of uniform density rolls up a curved surface with an initial velocity $3\ \mathrm{m/s}$ (as shown in figure). Maximum height with respect to the initial position covered by it will be _____ cm. (Take $g = 10\ \mathrm{m/s^2}$)



Open solution

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Q234. JEE Main 2023 (Online) 8th April Morning Shift

The moment of inertia of a semicircular ring about an axis, passing through the center and perpendicular to the plane of ring, is $\frac{1}{x}\mathrm{MR}^2$, where R is the radius and M is the mass of the semicircular ring. The value of x will be _____.

Open solution

Q235. JEE Main 2023 (Online) 6th April Evening Shift

A ring and a solid sphere rotating about an axis passing through their centers have same radii of gyration. The axis of rotation is perpendicular to plane of ring. The ratio of radius of ring to that of sphere is $\sqrt{\frac{2}{x}}$. The value of x is _____.

Open solution

Q236. JEE Main 2023 (Online) 6th April Morning Shift

Two identical solid spheres each of mass $2\ \mathrm{kg}$ and radii $10\ \mathrm{cm}$ are fixed at the ends of a light rod. The separation between the centres of the spheres is $40\ \mathrm{cm}$. The moment of inertia of the system about an axis perpendicular to the rod passing through its middle point is _____ $\times 10^{-3}\ \mathrm{kg\,m^2}$

Open solution

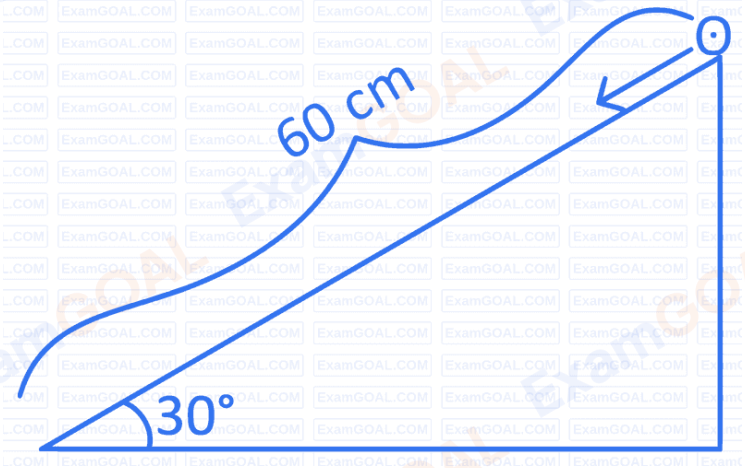
Q237. JEE Main 2023 (Online) 1st February Evening Shift

Moment of inertia of a disc of mass 'M' and radius 'R' about any of its diameter is $\frac{MR^2}{4}$. The moment of inertia of this disc about an axis normal to the disc and passing through a point on its edge will be, $\frac{x}{2}$ MR^2. The value of x is _____.

Open solution

Q238. JEE Main 2023 (Online) 1st February Morning Shift

A solid cylinder is released from rest from the top of an inclined plane of inclination 30° and length $60\ \mathrm{cm}$. If the cylinder rolls without slipping, its speed upon reaching the bottom of the inclined plane is _____ ms^{-1}. (Given $\mathrm{g} = 10\ \mathrm{ms}^{-2}$)



Open solution

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Q239. JEE Main 2023 (Online) 31st January Evening Shift

Two discs of same mass and different radii are made of different materials such that their thicknesses are $1\ \text{cm}$ and $0.5\ \text{cm}$ respectively. The densities of materials are in the ratio $3:5$. The moment of inertia of these discs respectively about their diameters will be in the ratio of $\frac{x}{6}$. The value of x is _____.

Open solution

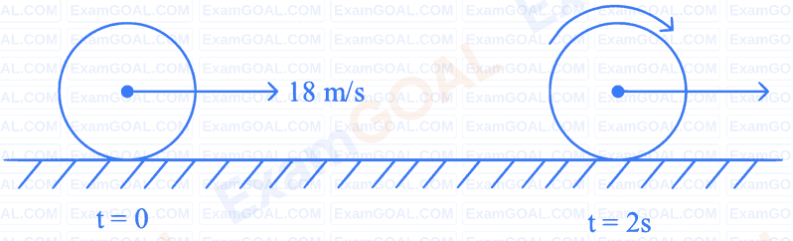
Q240. JEE Main 2023 (Online) 31st January Morning Shift

A solid sphere of mass $1\ \text{kg}$ rolls without slipping on a plane surface. Its kinetic energy is $7 \times 10^{-3}\ \text{J}$. The speed of the centre of mass of the sphere is _____ cm s^{-1}

Open solution

Q241. JEE Main 2023 (Online) 30th January Evening Shift

A uniform disc of mass $0.5\ \text{kg}$ and radius r is projected with velocity $18\ \text{m/s}$ at $\text{t} = 0\ \text{s}$ on a rough horizontal surface. It starts off with a purely sliding motion at $\text{t} = 0\ \text{s}$. After $2\ \text{s}$ it acquires a purely rolling motion (see figure). The total kinetic energy of the disc after $2\ \text{s}$ will be _____ J (given, coefficient of friction is 0.3 and $g = 10\ \text{m/s}^2$).



Open solution

Q242. JEE Main 2023 (Online) 30th January Morning Shift

A thin uniform rod of length $2\ \text{m}$, cross sectional area 'A' and density 'd' is rotated about an axis passing through the centre and perpendicular to its length with angular velocity ω. If value of ω in terms of its rotational kinetic energy E is $\sqrt{\frac{\alpha E}{Ad}}$ then value of α is _____.

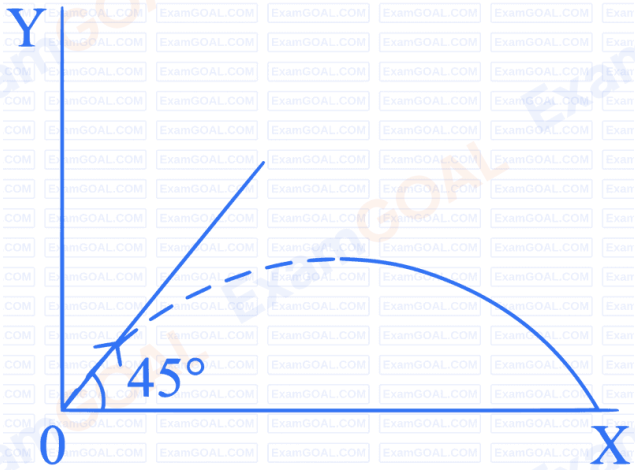
Open solution

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Q243. JEE Main 2023 (Online) 29th January Evening Shift

A particle of mass 100 g is projected at time t = 0 with a speed 20 ms^{-1} at an angle 45° to the horizontal as given in the figure. The magnitude of the angular momentum of the particle about the starting point at time t = 2s is found to be $\sqrt{K}\,kg\,m^2/s$. The value of K is _____.

(Take g = 10 ms^{-2})



Open solution

Q244. JEE Main 2023 (Online) 29th January Morning Shift

A solid sphere of mass 2 kg is making pure rolling on a horizontal surface with kinetic energy 2240 J. The velocity of centre of mass of the sphere will be _____ ms^{-1}.

Open solution

Q245. JEE Main 2023 (Online) 25th January Evening Shift

If a solid sphere of mass 5 kg and a disc of mass 4 kg have the same radius. Then the ratio of moment of inertia of the disc about a tangent in its plane to the moment of inertia of the sphere about its tangent will be $\frac{x}{7}$. The value of x is _____.

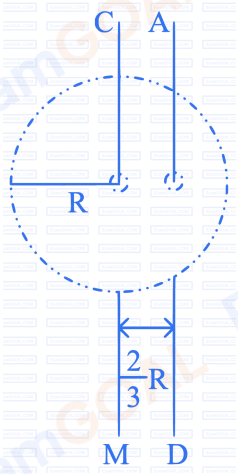
Open solution

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Q246. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

I_{CM} is the moment of inertia of a circular disc about an axis (CM) passing through its center and perpendicular to the plane of disc. I_{AB} is its moment of inertia about an axis AB perpendicular to plane and parallel to axis CM at a distance $\frac{2}{3}$R from center. Where R is the radius of the disc. The ratio of I_{AB} and I_{CM} is $x : 9$. The value of x is _____.



Q247. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

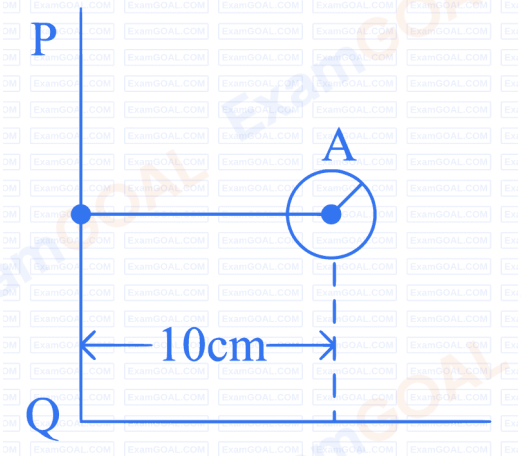
A uniform solid cylinder with radius R and length L has moment of inertia I_1, about the axis of the cylinder. A concentric solid cylinder of radius $R' = \frac{R}{2}$ and length $L' = \frac{L}{2}$ is carved out of the original cylinder. If I_2 is the moment of inertia of the carved out portion of the cylinder then $\frac{I_1}{I_2}$ = _____.

(Both I_1 and I_2 are about the axis of the cylinder)

Q248. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Solid sphere A is rotating about an axis PQ. If the radius of the sphere is 5 cm then its radius of gyration about PQ will be $\sqrt{x}$ cm. The value of x is _____.



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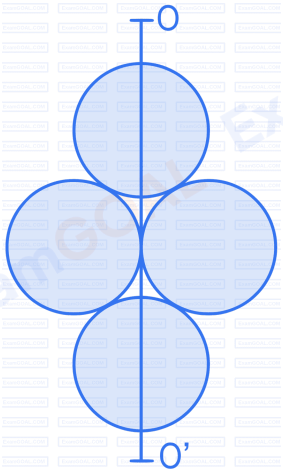
Q249. JEE Main 2022 (Online) 30th July Evening Shift

A fly wheel of mass $10\ \mathrm{kg}$ and radius $50\ \mathrm{cm}$ is rotating at a rate of 360 rpm. Assuming mass to be concentrated at the rim, the constant retarding torque required to stop the rotation of wheel in 6 rotations will be $x\pi\ \mathrm{N\,m}$. The value of x is _____

Open solution

Q250. JEE Main 2022 (Online) 28th July Morning Shift

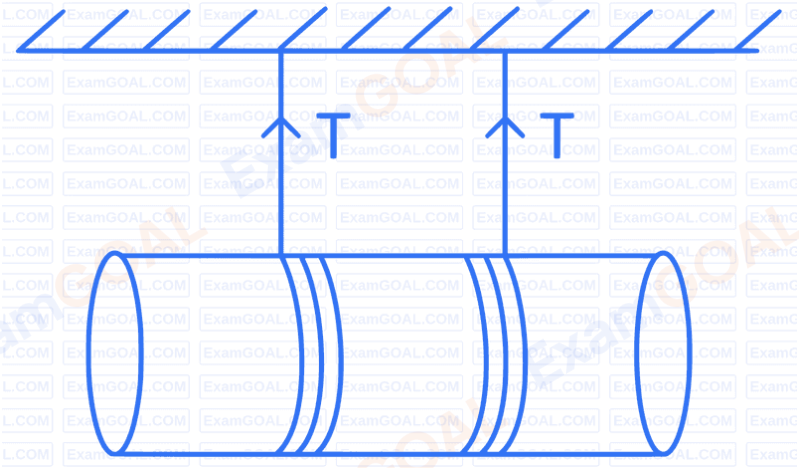
Four identical discs each of mass 'M' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}\ \mathrm{Ma}^2$. Then, the value of x will be _____.



Open solution

Q251. JEE Main 2022 (Online) 27th July Evening Shift

A solid cylinder is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should be unbinding the strings to achieve a speed of $4\ \mathrm{ms}^{-1}$, is _____ cm. (take g = $10\ \mathrm{ms}^{-2}$)



Open solution

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Q252. JEE Main 2022 (Online) 27th July Morning Shift

A pulley of radius $1.5\ \mathrm{m}$ is rotated about its axis by a force $F = (12\mathrm{t} - 3\mathrm{t}^2)N$ applied tangentially (while t is measured in seconds). If moment of inertia of the pulley about its axis of rotation is $4.5\ \mathrm{kg\,m}^2$, the number of rotations made by the pulley before its direction of motion is reversed, will be $\frac{K}{\pi}$. The value of K is _____.

Open solution

Q253. JEE Main 2022 (Online) 26th July Evening Shift

The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the center will be _____ m.

Given, the length of the rod is $10\sqrt{3}\ \mathrm{m}$.

Open solution

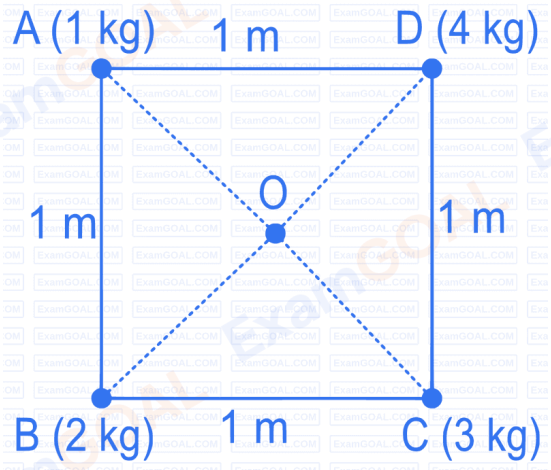
Q254. JEE Main 2022 (Online) 26th July Morning Shift

A disc of mass $1\ \mathrm{kg}$ and radius R is free to rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be $4\sqrt{\frac{x}{3R}}\ \mathrm{rad}\ s^{-1}$ where $x =$ _____. $\left(g = 10\ \mathrm{ms}^{-2}\right)$

Open solution

Q255. JEE Main 2022 (Online) 30th June Morning Shift

Four particles with a mass of 1 kg, 2 kg, 3 kg and 4 kg are situated at the corners of a square with side 1 m (as shown in the figure). The moment of inertia of the system, about an axis passing through the point O and perpendicular to the plane of the square, is _____ kg m^2.



Open solution

Q256. JEE Main 2022 (Online) 29th June Evening Shift

The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1. The same rod is bent into a ring and its moment of inertia about a diameter is I_2. If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be _____.

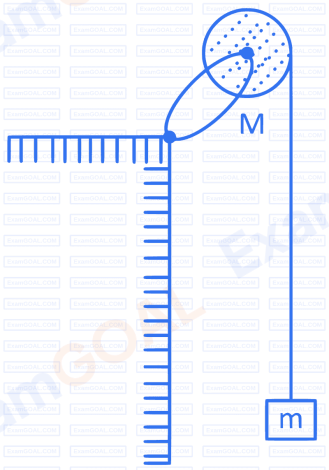
Open solution

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Q257. JEE Main 2022 (Online) 28th June Evening Shift

A uniform disc with mass M = 4 kg and radius R = 10 cm is mounted on a fixed horizontal axle as shown in figure. A block with mass m = 2 kg hangs from a massless cord that is wrapped around the rim of the disc. During the fall of the block, the cord does not slip and there is no friction at the axle. The tension in the cord is _____ N.

(Take g = 10 ms^{-2})



Open solution

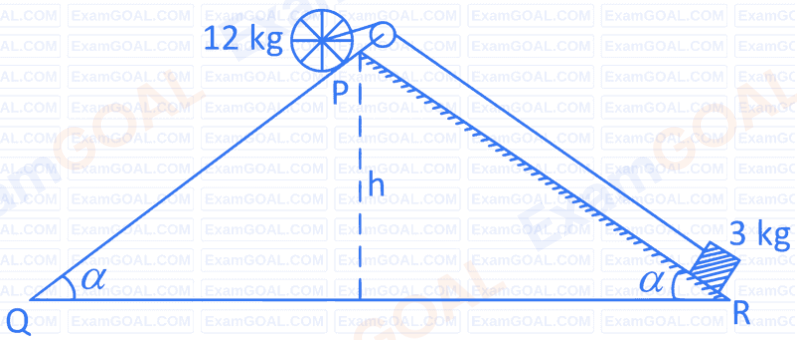
Q258. JEE Main 2022 (Online) 28th June Morning Shift

The position vector of 1 kg object is $\vec{r} = \left(3\hat{i} - \hat{j}\right)m$ and its velocity $\vec{v} = \left(3\hat{j} + \hat{k}\right)ms^{-1}$. The magnitude of its angular momentum is $\sqrt{x}$ Nm where x is _____.

Open solution

Q259. JEE Main 2022 (Online) 27th June Evening Shift

A rolling wheel of 12 kg is on an inclined plane at position P and connected to a mass of 3 kg through a string of fixed length and pulley as shown in figure. Consider PR as friction free surface. The velocity of centre of mass of the wheel when it reaches at the bottom Q of the inclined plane PQ will be $\frac{1}{2}\sqrt{xgh}$ m/s. The value of x is _____.



Open solution

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Q260. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

Moment of Inertia (M.I.) of four bodies having same mass 'M' and radius '2R' are as follows:

I_1 = M.I. of solid sphere about its diameter

I_2 = M.I. of solid cylinder about its axis

I_3 = M.I. of solid circular disc about its diameter

I_4 = M.I. of thin circular ring about its diameter

If $2(I_2 + I_3) + I_4 = x \, . \, I_1$, then the value of x will be _____.

Q261. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

A metre scale is balanced on a knife edge at its centre. When two coins, each of mass 10 g are put one on the top of the other at the 10.0 cm mark the scale is found to be balanced at 40.0 cm mark. The mass of the metre scale is found to be $x \times 10^{-2}$ kg. The value of x is _____.

Q262. JEE Main 2021 (Online) 1st September Evening Shift

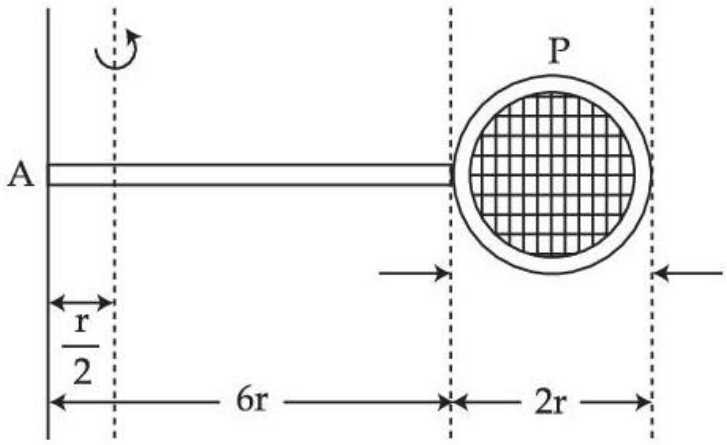
Open solution

A 2 kg steel rod of length 0.6 m is clamped on a table vertically at its lower end and is free to rotate in vertical plane. The upper end is pushed so that the rod falls under gravity, ignoring the friction due to clamping at its lower end, the speed of the free end of rod when it passes through its lowest position is _____ ms^{-1}. (Take g = 10 ms^{-2})

Q263. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Consider a badminton racket with length scales as shown in the figure.



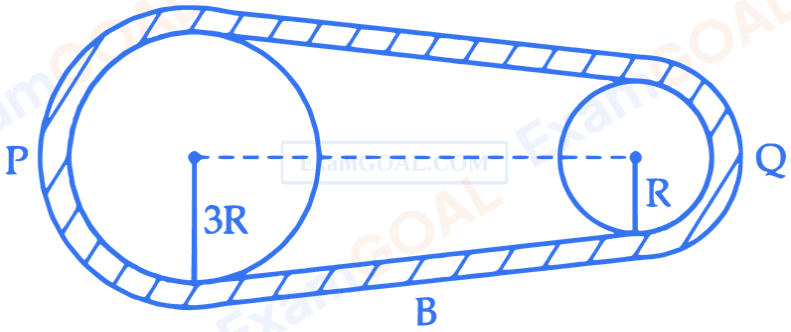
If the mass of the linear and circular portions of the badminton racket are same (M) and the mass of the threads are negligible, the moment of inertia of the racket about an axis perpendicular to the handle and in the plane of the ring at $\frac{r}{2}$ distance from the end A of the handle will be ..... Mr^2.

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Q264. JEE Main 2021 (Online) 27th July Evening Shift

In the given figure, two wheels P and Q are connected by a belt B. The radius of P is three times as that of Q. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be x : 1. The value of x will be _____.

Open solution



Q265. JEE Main 2021 (Online) 25th July Evening Shift

A solid disc of radius 20 cm and mass 10 kg is rotating with an angular velocity of 600 rpm, about an axis normal to its circular plane and passing through its centre of mass. The retarding torque required to bring the disc at rest in 10 s is _____ $\pi \times 10^{-1}$ Nm.

Open solution

Q266. JEE Main 2021 (Online) 25th July Morning Shift

A particle of mass 'm' is moving in time 't' on a trajectory given by

$$\vec{r} = 10\alpha t^2\hat{i} + 5\beta(t-5)\hat{j}$$

Where α and β are dimensional constants.

The angular momentum of the particle becomes the same as it was for t = 0 at time t = _____ seconds.

Open solution

Q267. JEE Main 2021 (Online) 22th July Evening Shift

The centre of a wheel rolling on a plane surface moves with a speed v_0. A particle on the rim of the wheel at the same level as the centre will be moving at a speed $\sqrt{x}v_0$. Then the value of x is _____.

Open solution

Q268. JEE Main 2021 (Online) 20th July Evening Shift

Two bodies, a ring and a solid cylinder of same material are rolling down without slipping an inclined plane. The radii of the bodies are same. The ratio of velocity of the centre of mass at the bottom of the inclined plane of the ring to that of the cylinder is $\frac{\sqrt{x}}{2}$. Then, the value of x is _____.

Open solution

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Q269. JEE Main 2021 (Online) 20th July Evening Shift

A body rotating with an angular speed of 600 rpm is uniformly accelerated to 1800 rpm in 10 sec. The number of rotations made in the process is _____.

Open solution

Q270. JEE Main 2021 (Online) 20th July Morning Shift

A circular disc reaches from top to bottom of an inclined plane of length 'L'. When it slips down the plane, it makes time 't_1'. When it rolls down the plane, it takes time t_2. The value of $\frac{t_2}{t_1}$ is $\sqrt{\frac{3}{x}}$. The value of x will be _____.

Open solution

Q271. JEE Main 2021 (Online) 17th March Morning Shift

The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is _____. (Assuming the acceleration to be uniform).

Open solution

Q272. JEE Main 2021 (Online) 17th March Morning Shift

The following bodies,

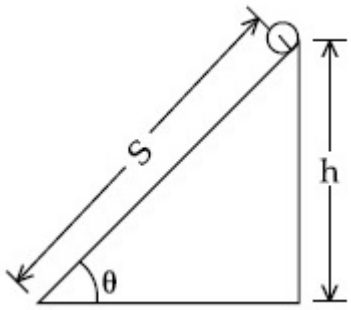
(1) a ring

(2) a disc

(3) a solid cylinder

(4) a solid sphere,

of same mass 'm' and radius 'R' are allowed to roll down without slipping simultaneously from the top of the inclined plane. The body which will reach first at the bottom of the inclined plane is _____. [Mark the body as per their respective numbering given in the question]



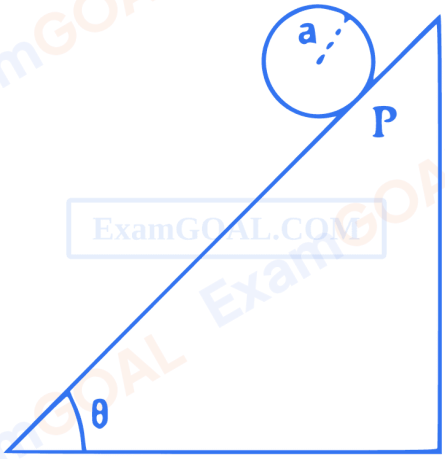
Open solution

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Q273. JEE Main 2021 (Online) 16th March Evening Shift

A solid disc of radius 'a' and mass 'm' rolls down without slipping on an inclined plane making an angle θ with the horizontal. The acceleration of the disc will be $\frac{2}{b}$g sinθ where b is _____. (Round off to the Nearest Integer) (g = acceleration due to gravity, θ = angle as shown in figure)

Open solution



Q274. JEE Main 2021 (Online) 16th March Evening Shift

A force $\vec{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ is applied on an intersection point of x = 2 plane and x-axis. The magnitude of torque of this force about a point (2, 3, 4) is _____. (Round off to the Nearest Integer)

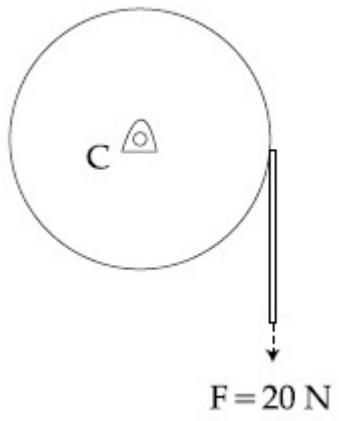
Open solution

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Q275. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

Consider a 20 kg uniform circular disk of radius 0.2 m. It is pin supported at its center and is at rest initially. The disk is acted upon by a constant force F = 20 N through a massless string wrapped around its periphery as shown in the figure.



Suppose the disk makes n number of revolutions to attain an angular speed of 50 rad s^{-1}.

The value of n, to the nearest integer, is _____.

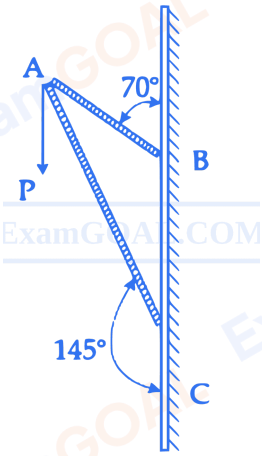
[Given : In one complete revolution, the disk rotates by 6.28 rad]

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Q276. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

Consider a frame that is made up of two thin massless rods AB and AC as shown in the figure. A vertical force $\vec{P}$ of magnitude 100 N is applied at point A of the frame.



Suppose the force $\vec{P}$ is resolved parallel to the arms AB and AC of the frame.

The magnitude of the resolved component along the arm AC is xN.

The value of x, to the nearest integer, is _____.

[Given : sin(35°) = 0.573, cos(35°) = 0.819

sin(110°) = 0.939, cos(110°) = $-$ 0.342]

Q277. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

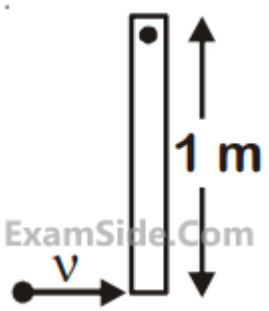
A uniform thin bar of mass 6 kg and length 2.4 meter is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is _____ $\times 10^{-1}$ kg m^2.

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Q278. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

A thin rod of mass 0.9 kg and length 1 m is suspended, at rest, from one end so that it can freely oscillate in the vertical plane. A particle of mass 0.1 kg moving in a straight line with velocity 80 m/s hits the rod at its bottom most point and sticks to it (see figure). The angular speed (in rad/s) of the rod immediately after the collision will be _____.



Q279. JEE Main 2020 (Online) 5th September Morning Slot

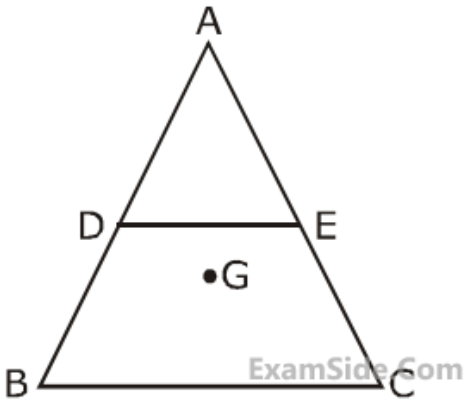
Open solution

A force $\vec{F} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right)$ N acts at a point $\left(4\hat{i} + 3\hat{j} - \hat{k}\right)$ m. Then the magnitude of torque about the point $\left(\hat{i} + 2\hat{j} + \hat{k}\right)$ m will be $\sqrt{x}$ N m. The value of x is _____.

Q280. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

ABC is a plane lamina of the shape of an equilateral triangle. D, E are mid points of AB, AC and G is the centroid of the lamina. Moment of inertia of the lamina about an axis passing through G and perpendicular to the plane ABC is I_0 . If part ADE is removed, the moment of inertia of the remaining part about the same axis is $\frac{NI_0}{16}$ where N is an integer. Value of N is _____.



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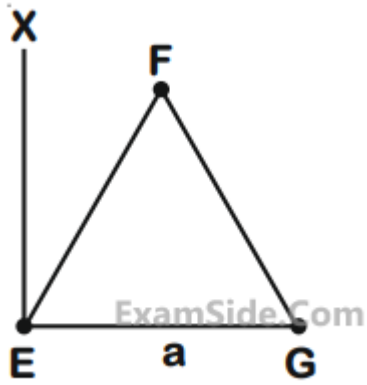
Q281. JEE Main 2020 (Online) 4th September Morning Slot

A circular disc of mass M and radius R is rotating about its axis with angular speed ω_1 . If another stationary disc having radius $\frac{R}{2}$ and same mass M is dropped co-axially on to the rotating disc. Gradually both discs attain constant angular speed ω_2 the energy lost in the process is p% of the initial energy. Value of p is _____.

Open solution

Q282. JEE Main 2020 (Online) 3rd September Evening Slot

A massless equilateral triangle EFG of side 'a' (As shown in figure) has three particles of mass m situated at its vertices. The moment of inertia of the system about the line EX perpendicular to EG in the plane of EFG is $\frac{N}{20}$ ma^2 where N is an integer. The value of N is _____.



Open solution

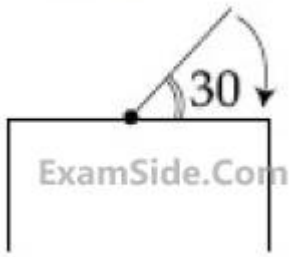
Q283. JEE Main 2020 (Online) 3rd September Morning Slot

A person of 80 kg mass is standing on the rim of a circular platform of mass 200 kg rotating about its axis at 5 revolutions per minute (rpm). The person now starts moving towards the centre of the platform. What will be the rotational speed (in rpm) of the platform when the person reaches its centre _____.

Open solution

Q284. JEE Main 2020 (Online) 9th January Morning Slot

One end of a straight uniform 1m long bar is pivoted on horizontal table. It is released from rest when it makes an angle 30° from the horizontal (see figure). Its angular speed when it hits the table is given as $\sqrt{n}$ s^{-1}, where n is an integer. The value of n is _____.

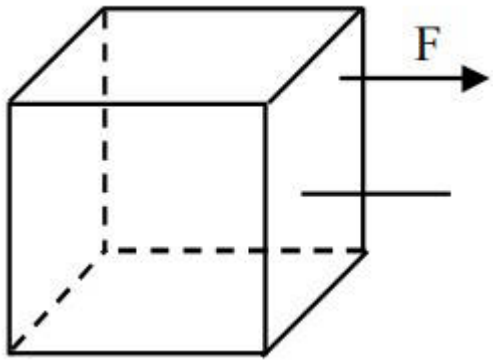


Open solution

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Q285. JEE Main 2020 (Online) 7th January Evening Slot

Open solution



Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is μ = 0.4, the maximum possible value of 100 × $\frac{b}{a}$ for box not to topple before moving is .....

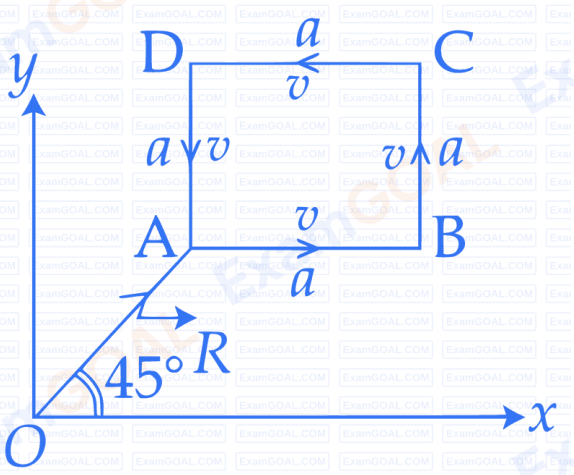
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MCQ (More than One Correct Answer) 1 questions

Q286. JEE Main 2016 (Offline)

A particle of mass m is moving along the side of a square of side 'a', with a uniform speed v in the x-y plane as shown in the figure :

Open solution



Which of the following statements is **false** for the angular momentum $\vec{L}$ about the origin ?

(A) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} + a\right]\hat{k}$
when the particle is moving from B to C.

(B) $\vec{L} = \frac{mv}{\sqrt{2}}R\hat{k}$
when the particle is moving from D to A.

(C) $\vec{L} = -\frac{mv}{\sqrt{2}}R\hat{k}$
when the particle is moving from A to B

(D) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} - a\right]\hat{k}$
when the particle is moving from C to D.

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **D**	4. **B**	5. **D**
6. **B**	7. **D**	8. **D**	9. **D**	10. **B**
11. **D**	12. **B**	13. **D**	14. **A**	15. **C**
16. **D**	17. **B**	18. **A**	19. **B**	20. **A**
21. **A**	22. **C**	23. **C**	24. **B**	25. **A**
26. **A**	27. **A**	28. **A**	29. **C**	30. **A**
31. **C**	32. **D**	33. **A**	34. **C**	35. **B**
36. **B**	37. **C**	38. **D**	39. **C**	40. **A**
41. **D**	42. **A**	43. **C**	44. **C**	45. **D**
46. **C**	47. **D**	48. **C**	49. **B**	50. **C**
51. **A**	52. **C**	53. **B**	54. **C**	55. **B**
56. **B**	57. **C**	58. **B**	59. **C**	60. **D**
61. **D**	62. **C**	63. **D**	64. **B**	65. **C**
66. **B**	67. **D**	68. **D**	69. **C**	70. **D**
71. **D**	72. **A**	73. **D**	74. **B**	75. **D**
76. **B**	77. **A**	78. **C**	79. **A**	80. **B**
81. **B**	82. **D**	83. **B**	84. **B**	85. **B**
86. **B**	87. **A**	88. **D**	89. **A**	90. **D**
91. **A**	92. **C**	93. **A**	94. **A**	95. **C**
96. **C**	97. **C**	98. **B**	99. **A**	100. **C**
101. **B**	102. **B**	103. **C**	104. **A**	105. **B**
106. **D**	107. **C**	108. **A**	109. **D**	110. **B**
111. **D**	112. **C**	113. **A**	114. **C**	115. **B**
116. **A**	117. **B**	118. **A**	119. **A**	120. **D**
121. **B**	122. **A**	123. **B**	124. **D**	125. **B**
126. **B**	127. **D**	128. **C**	129. **C**	130. **A**
131. **A**	132. **B**	133. **B**	134. **A**	135. **A**
136. **B**	137. **B**	138. **B**	139. **B**	140. **C**
141. **A**	142. **B**	143. **B**	144. **A**	145. **C**
146. **C**	147. **C**	148. **C**	149. **A**	150. **D**

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151. **B**	152. **A**	153. **D**	154. **C**	155. **B**
156. **C**	157. **A**	158. **A**	159. **B**	160. **C**
161. **D**	162. **B**	163. **D**	164. **B**	165. **D**
166. **B**	167. **B**	168. **C**	169. **A**	170. **C**
171. **C**	172. **D**	173. **B**	174. **D**	175. **D**
176. **C**	177. **D**	178. **B**	179. **C**	180. **C**
181. **B**	182. **A**	183. **D**	184. **A**	185. **D**
186. **C**	187. **D**	188. **D**	189. **A**	

Numerical

190. **15**	191. **30**	192. **265**	193. **162**	194. **109**
195. **2**	196. **16**	197. **17**	198. **40**	199. **9**
200. **199**	201. **32**	202. **4 or 3.5**	203. **1**	204. **2**
205. **16**	206. **4**	207. **90**	208. **3**	209. **100**
210. **2**	211. **4**	212. **13**	213. **2**	214. **7**
215. **4**	216. **8**	217. **53**	218. **6**	219. **24**
220. **250**	221. **60**	222. **10**	223. **7**	224. **16**
225. **5**	226. **15**	227. **4**	228. **2**	229. **3**
230. **35**	231. **3**	232. **16**	233. **75**	234. **1**
235. **5**	236. **176**	237. **3**	238. **2**	239. **5**
240. **10**	241. **54**	242. **3**	243. **800**	244. **40**
245. **5**	246. **17**	247. **32**	248. **110**	249. **15**
250. **3**	251. **120**	252. **18**	253. **5**	254. **5**
255. **5**	256. **8**	257. **10**	258. **91**	259. **3**
260. **5**	261. **6**	262. **6**	263. **52**	264. **9**
265. **4**	266. **10**	267. **02**	268. **3**	269. **200**
270. **2**	271. **728**	272. **4**	273. **3**	274. **20**
275. **20**	276. **82**	277. **8**	278. **20**	279. **195**
280. **11**	281. **20**	282. **25**	283. **9**	284. **15**
285. **50**				

MCQ (More than One Correct Answer)

286. **B, D**

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